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United States Patent [19]

Choquier et al.

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[45] **Date of Patent: Jun. 30, 1998**

[54] **SYSTEM FOR ON-LINE SERVICE IN WHICH GATEWAY COMPUTER USES SERVICE MAP WHICH INCLUDES LOADING CONDITION OF SERVERS BROADCASTED BY APPLICATION SERVERS FOR LOAD BALANCING**

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[73] Assignee: **Microsoft Corporation**, Redmond, Wash.

[21] Appl. No.: **472,807**

[22] Filed: **Jun. 7, 1995**

[51] **Int. Cl.**[6] **G06F 13/14**

[52] **U.S. Cl.** **395/200.53**; 395/200.33; 395/200.37; 395/200.53; 395/200.57; 395/200.48; 395/200.79; 395/675; 364/238; 364/281; 370/480

[58] **Field of Search** 395/187, 200, 395/650, 307, 200.53, 200.33, 200.37, 200.57, 200.48, 200.79, 675; 370/85, 480; 364/238, 281

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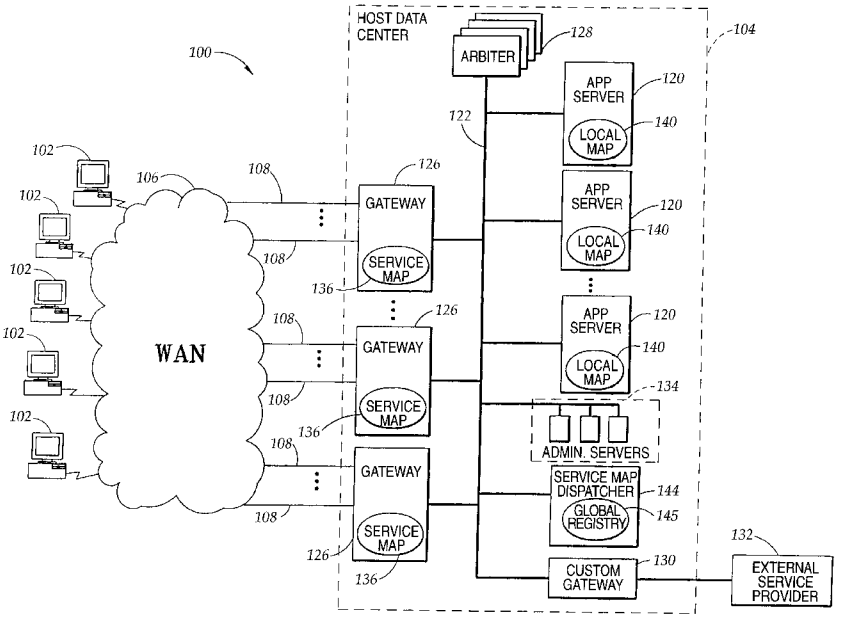
Primary Examiner—Thomas C. Lee
Assistant Examiner—Ki S. Kim
Attorney, Agent, or Firm—Leydig, Voit & Mayer, Ltd.

[57] **ABSTRACT**

An on-line services network includes application servers and Gateway microcomputers that are interconnected by a LAN. The Gateway microcomputers receive service requests which are transmitted over a WAN from client microcomputers operated by end users. Upon receiving a request to open a service, the Gateway microcomputers access a periodically-updated service map to locate the replicated application servers that are currently running the corresponding service application, and then apply a load balancing method (using server load data contained within the service map) to select an application server that has a relatively low processing load. A communications protocol between the client and Gateway microcomputers enables users to access multiple services simultaneously. A hot redirection technique allows a user service session to be transferred from one application server to another (replicated) application server without termination of the service session. Various mechanisms are provided for efficiently reallocating application servers to different on-line services, and for allowing additional servers to be efficiently added to the network.

59 Claims, 14 Drawing Sheets



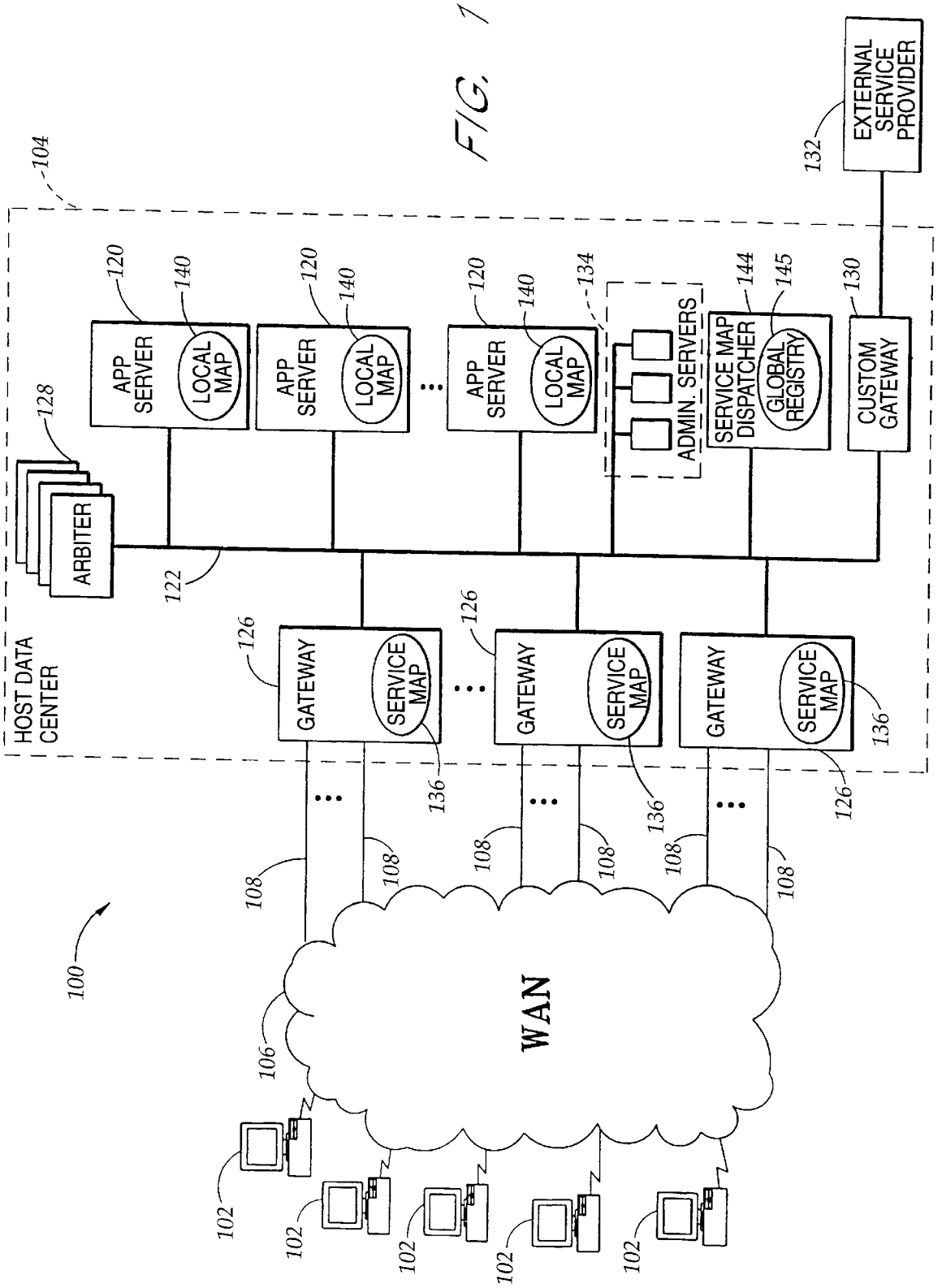

HOST DATA CENTER
104
ARBITER
128
APP SERVER
LOCAL MAP
120
140
122
ADMIN. SERVERS
134
SERVICE MAP DISPATCHER
GLOBAL REGISTRY
144
145
CUSTOM GATEWAY
130
EXTERNAL SERVICE PROVIDER
132
GATEWAY
SERVICE MAP
126
136
108
WAN
106
102
100

FIG. 1

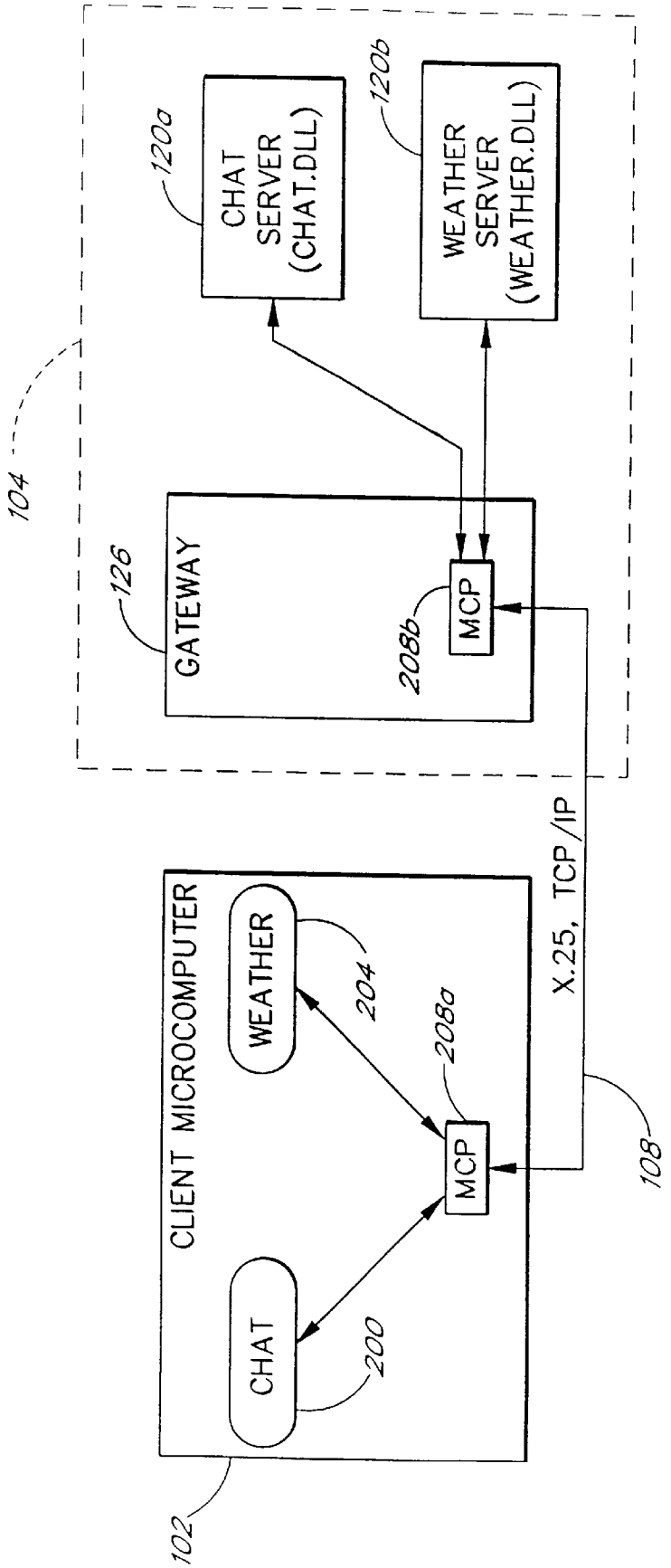

104
120a
CHAT SERVER (CHAT.DLL)
120b
WEATHER SERVER (WEATHER.DLL)
126
GATEWAY
208b
MCP
X.25, TCP/IP
108
CLIENT MICROCOMPUTER
WEATHER
204
208a
MCP
CHAT
200
102

FIG. 2

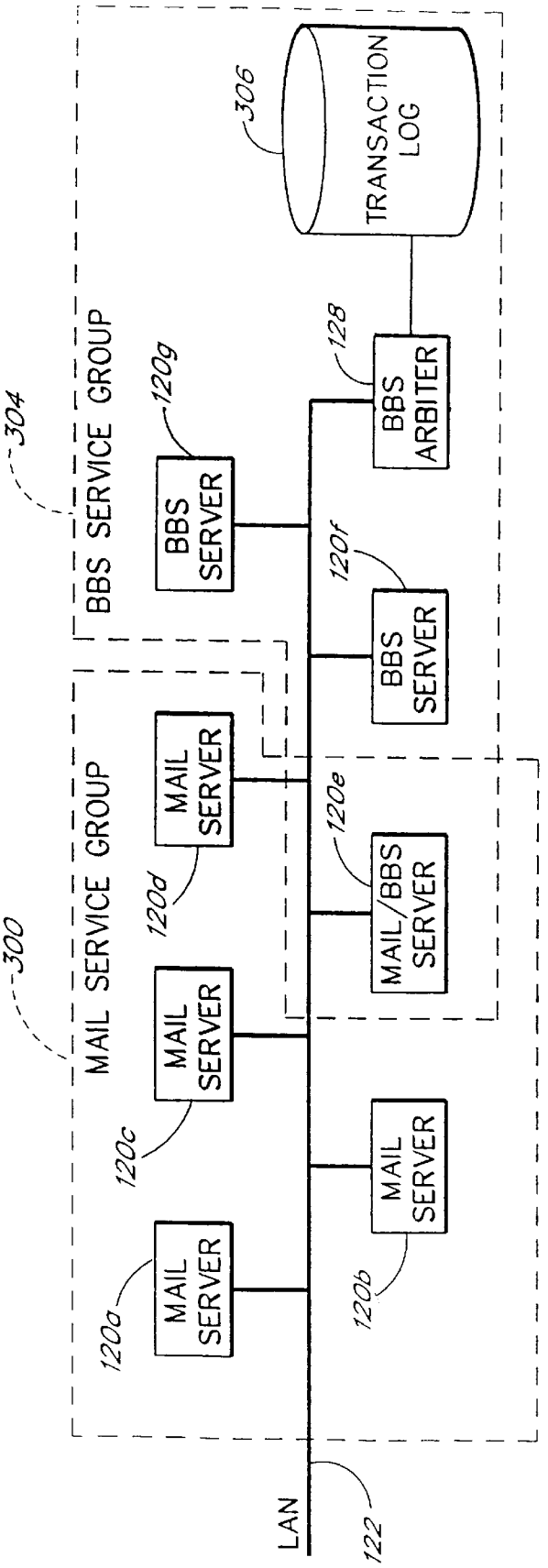

300
MAIL SERVICE GROUP
304
BBS SERVICE GROUP
306
TRANSACTION LOG
128
BBS ARBITER
120a
MAIL SERVER
120b
MAIL SERVER
120c
MAIL SERVER
120d
MAIL SERVER
120e
MAIL/BBS SERVER
120f
BBS SERVER
120g
BBS SERVER
LAN
122

FIG. 3

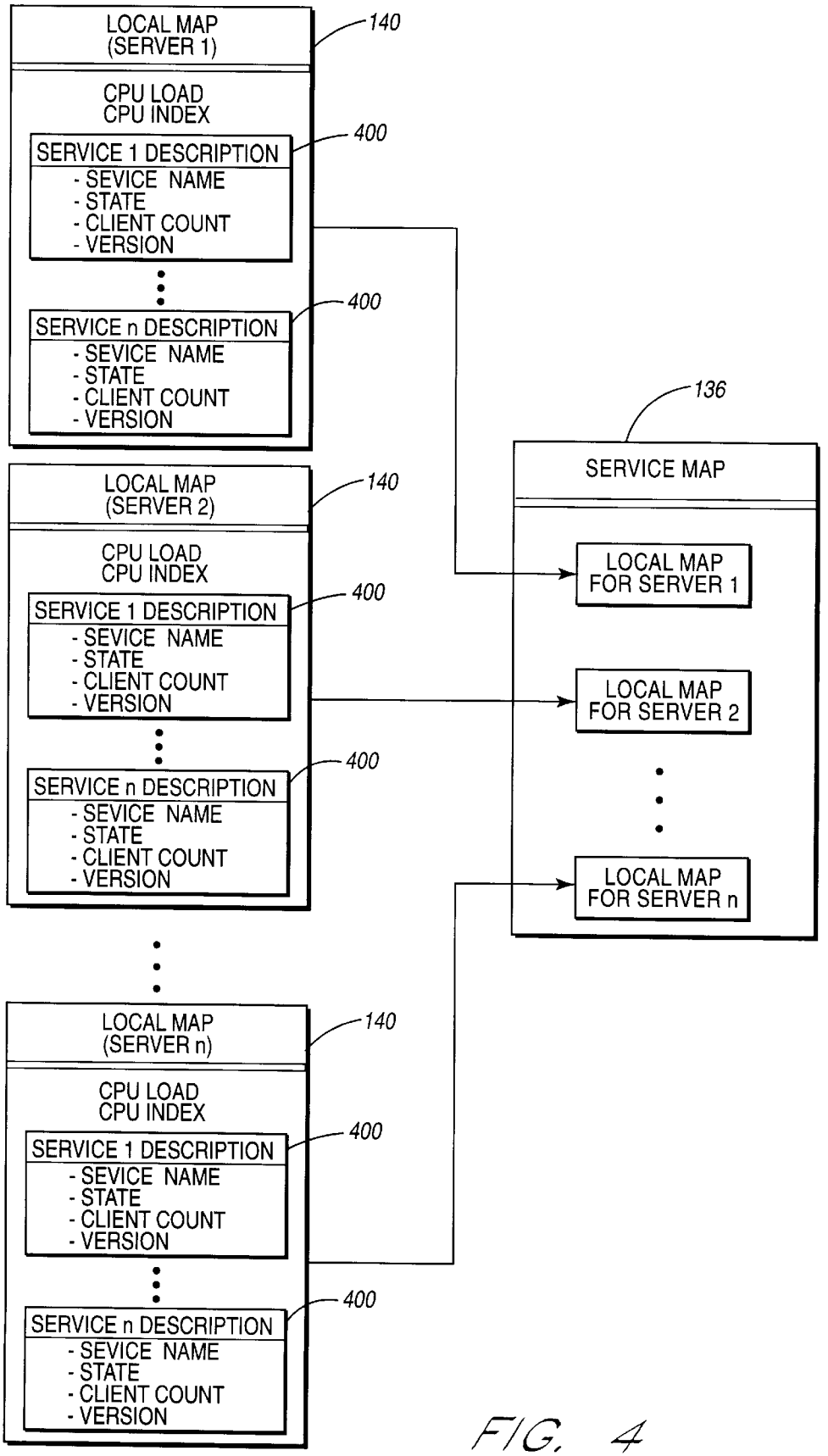

LOCAL MAP
(SERVER 1)
140
CPU LOAD
CPU INDEX
SERVICE 1 DESCRIPTION
400
- SEVICE NAME
- STATE
- CLIENT COUNT
- VERSION
SERVICE n DESCRIPTION
400
- SEVICE NAME
- STATE
- CLIENT COUNT
- VERSION
LOCAL MAP
(SERVER 2)
140
CPU LOAD
CPU INDEX
SERVICE 1 DESCRIPTION
400
- SEVICE NAME
- STATE
- CLIENT COUNT
- VERSION
SERVICE n DESCRIPTION
400
- SEVICE NAME
- STATE
- CLIENT COUNT
- VERSION
LOCAL MAP
(SERVER n)
140
CPU LOAD
CPU INDEX
SERVICE 1 DESCRIPTION
400
- SEVICE NAME
- STATE
- CLIENT COUNT
- VERSION
SERVICE n DESCRIPTION
400
- SEVICE NAME
- STATE
- CLIENT COUNT
- VERSION
136
SERVICE MAP
LOCAL MAP
FOR SERVER 1
LOCAL MAP
FOR SERVER 2
LOCAL MAP
FOR SERVER n

FIG. 4

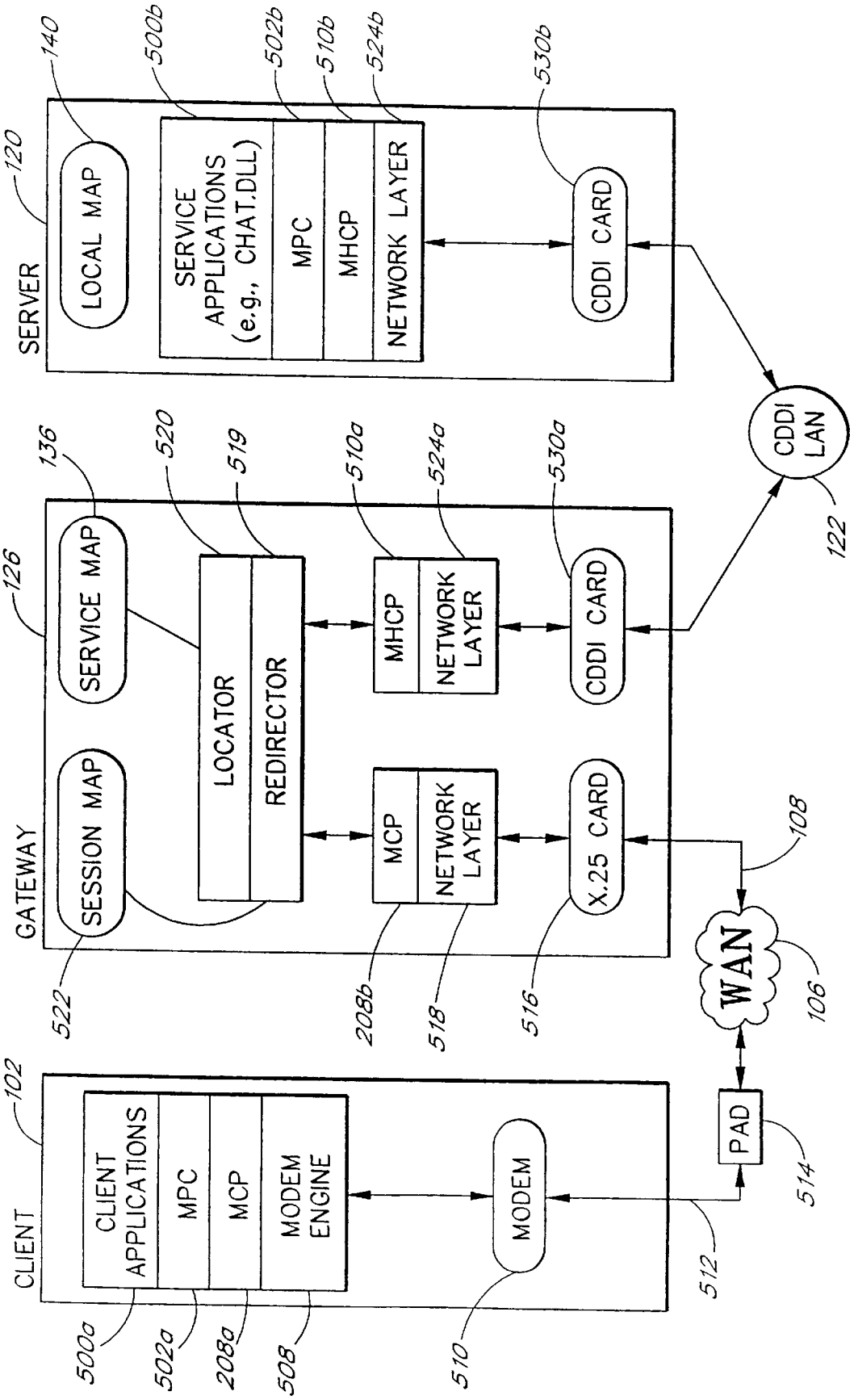

SERVER
120
LOCAL MAP
140
SERVICE APPLICATIONS (e.g., CHAT.DLL)
500b
MPC
502b
MHCP
510b
NETWORK LAYER
524b
CDDI CARD
530b
CDDI LAN
122
GATEWAY
126
SERVICE MAP
136
SESSION MAP
522
LOCATOR
520
REDIRECTOR
519
MHCP
510a
NETWORK LAYER
524a
CDDI CARD
530a
MCP
208b
NETWORK LAYER
518
X.25 CARD
516
108
WAN
106
CLIENT
102
CLIENT APPLICATIONS
500a
MPC
502a
MCP
208a
MODEM ENGINE
508
MODEM
510
512
PAD
514

FIG. 5A

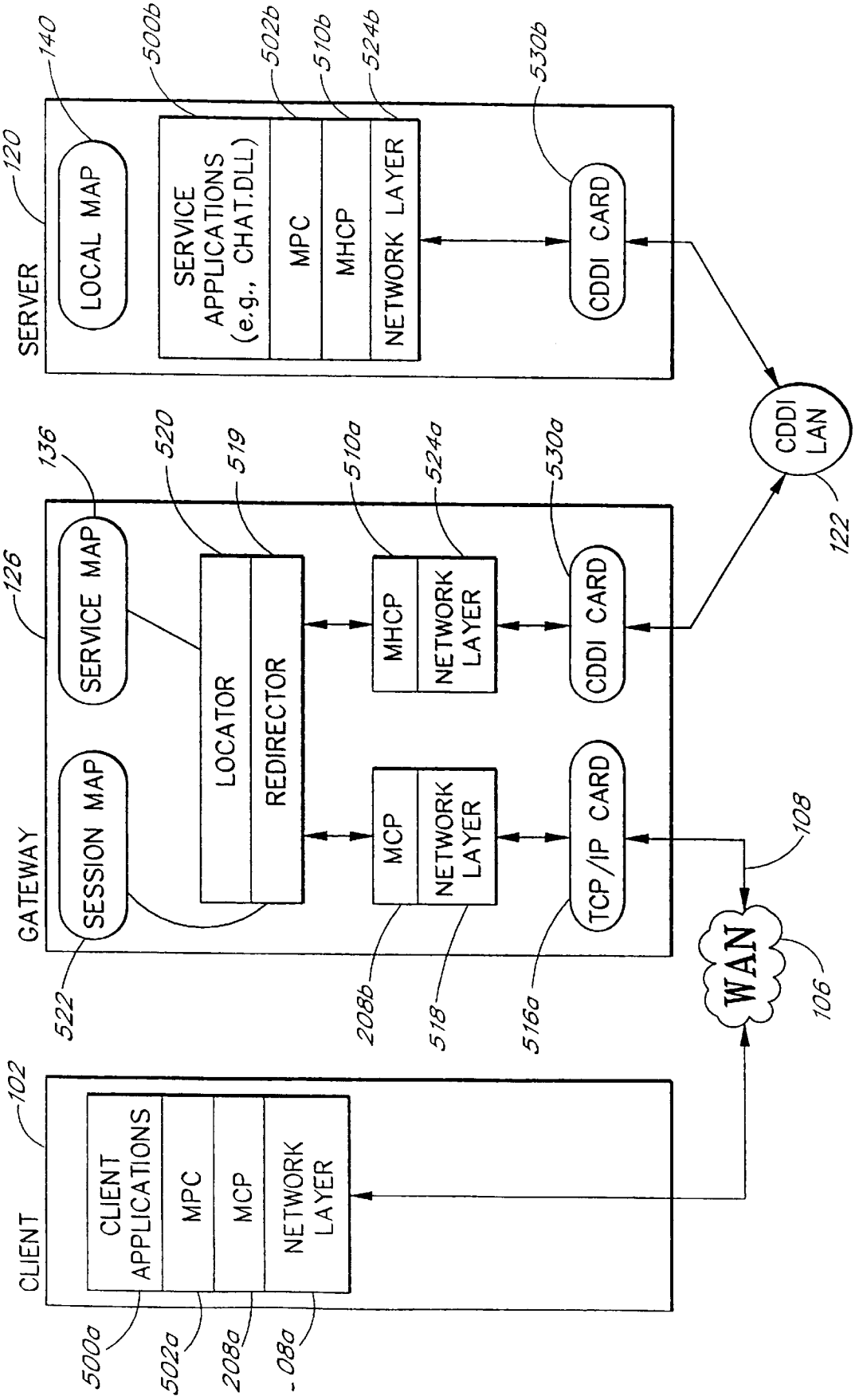

SERVER
120
LOCAL MAP
140
SERVICE APPLICATIONS (e.g., CHAT.DLL)
500b
MPC
502b
MHCP
510b
NETWORK LAYER
524b
CDDI CARD
530b
GATEWAY
126
SERVICE MAP
136
SESSION MAP
522
LOCATOR
520
REDIRECTOR
519
MHCP
510a
NETWORK LAYER
524a
MCP
208b
NETWORK LAYER
518
CDDI CARD
530a
TCP/IP CARD
516a
CDDI LAN
122
WAN
106
108
CLIENT
102
CLIENT APPLICATIONS
500a
MPC
502a
MCP
208a
NETWORK LAYER
108a

FIG. 5B

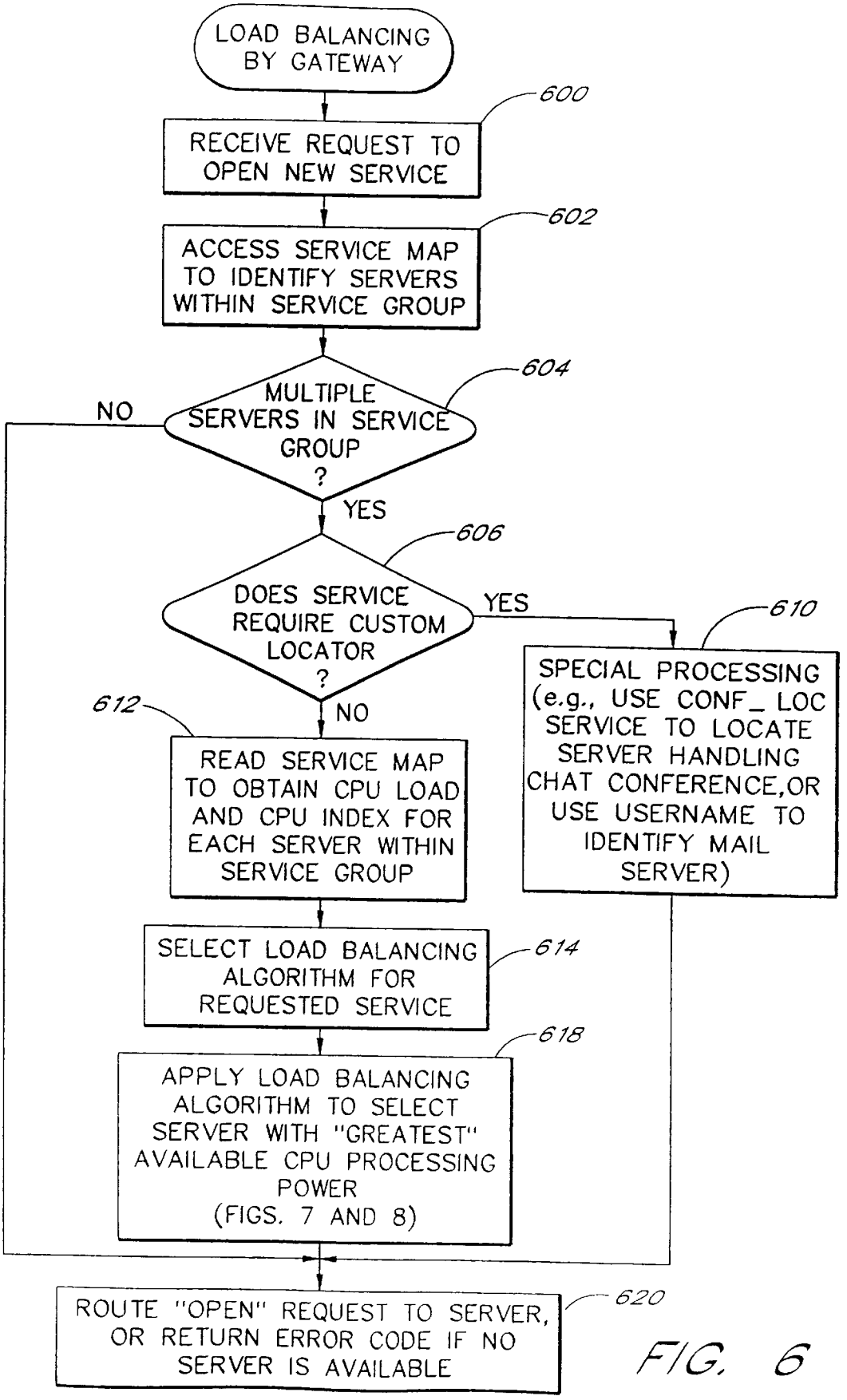

LOAD BALANCING BY GATEWAY
600
RECEIVE REQUEST TO OPEN NEW SERVICE
602
ACCESS SERVICE MAP TO IDENTIFY SERVERS WITHIN SERVICE GROUP
604
MULTIPLE SERVERS IN SERVICE GROUP ?
NO
YES
606
DOES SERVICE REQUIRE CUSTOM LOCATOR ?
YES
610
SPECIAL PROCESSING (e.g., USE CONF_ LOC SERVICE TO LOCATE SERVER HANDLING CHAT CONFERENCE,OR USE USERNAME TO IDENTIFY MAIL SERVER)
NO
612
READ SERVICE MAP TO OBTAIN CPU LOAD AND CPU INDEX FOR EACH SERVER WITHIN SERVICE GROUP
SELECT LOAD BALANCING ALGORITHM FOR REQUESTED SERVICE
614
618
APPLY LOAD BALANCING ALGORITHM TO SELECT SERVER WITH "GREATEST" AVAILABLE CPU PROCESSING POWER (FIGS. 7 AND 8)
ROUTE "OPEN" REQUEST TO SERVER, OR RETURN ERROR CODE IF NO SERVER IS AVAILABLE
620

FIG. 6

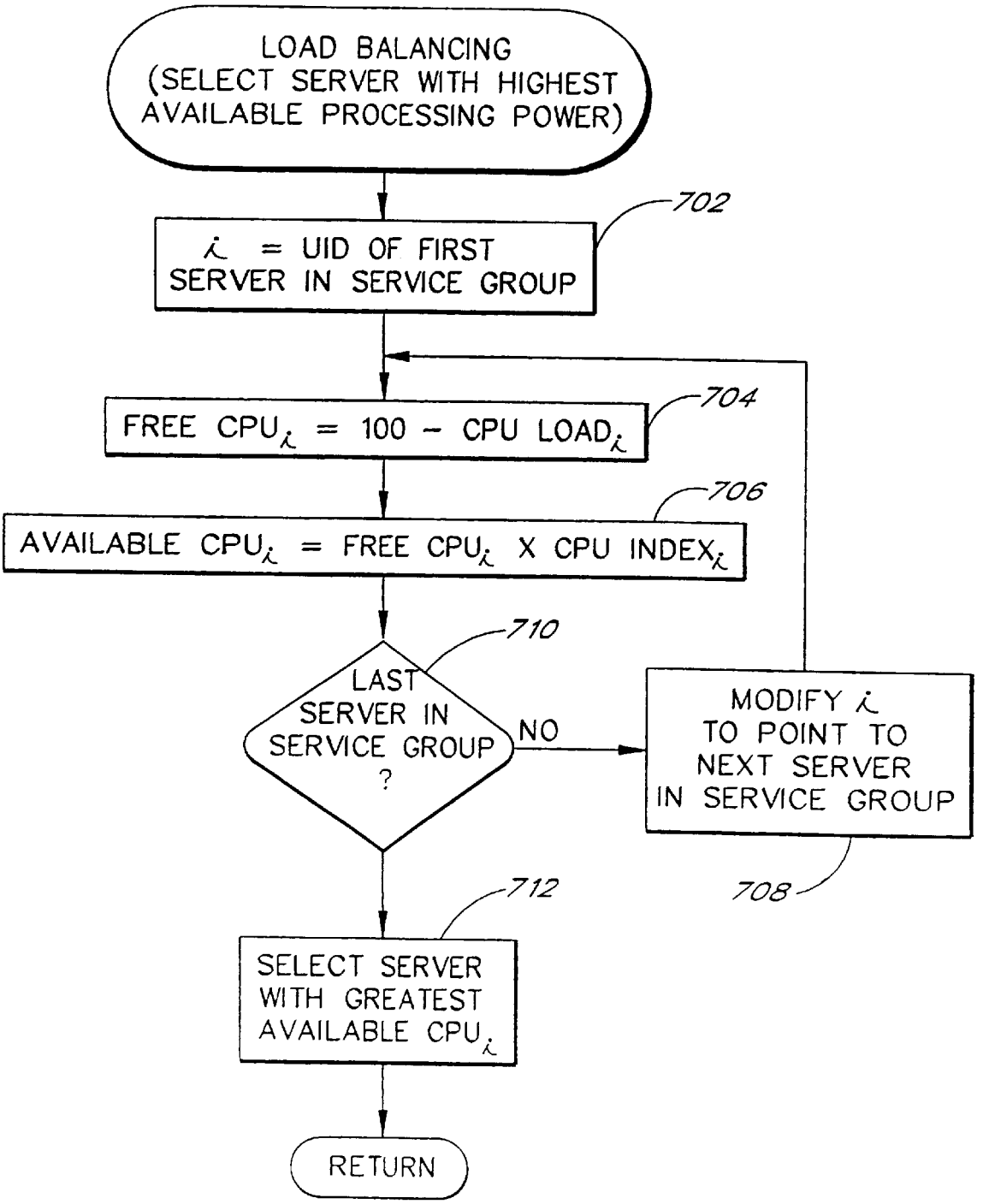

LOAD BALANCING
(SELECT SERVER WITH HIGHEST
AVAILABLE PROCESSING POWER)
i = UID OF FIRST
SERVER IN SERVICE GROUP
702
FREE CPU$_i$ = 100 − CPU LOAD$_i$
704
AVAILABLE CPU$_i$ = FREE CPU$_i$ X CPU INDEX$_i$
706
LAST
SERVER IN
SERVICE GROUP
?
710
NO
MODIFY i
TO POINT TO
NEXT SERVER
IN SERVICE GROUP
708
SELECT SERVER
WITH GREATEST
AVAILABLE CPU$_i$
712
RETURN

FIG. 7

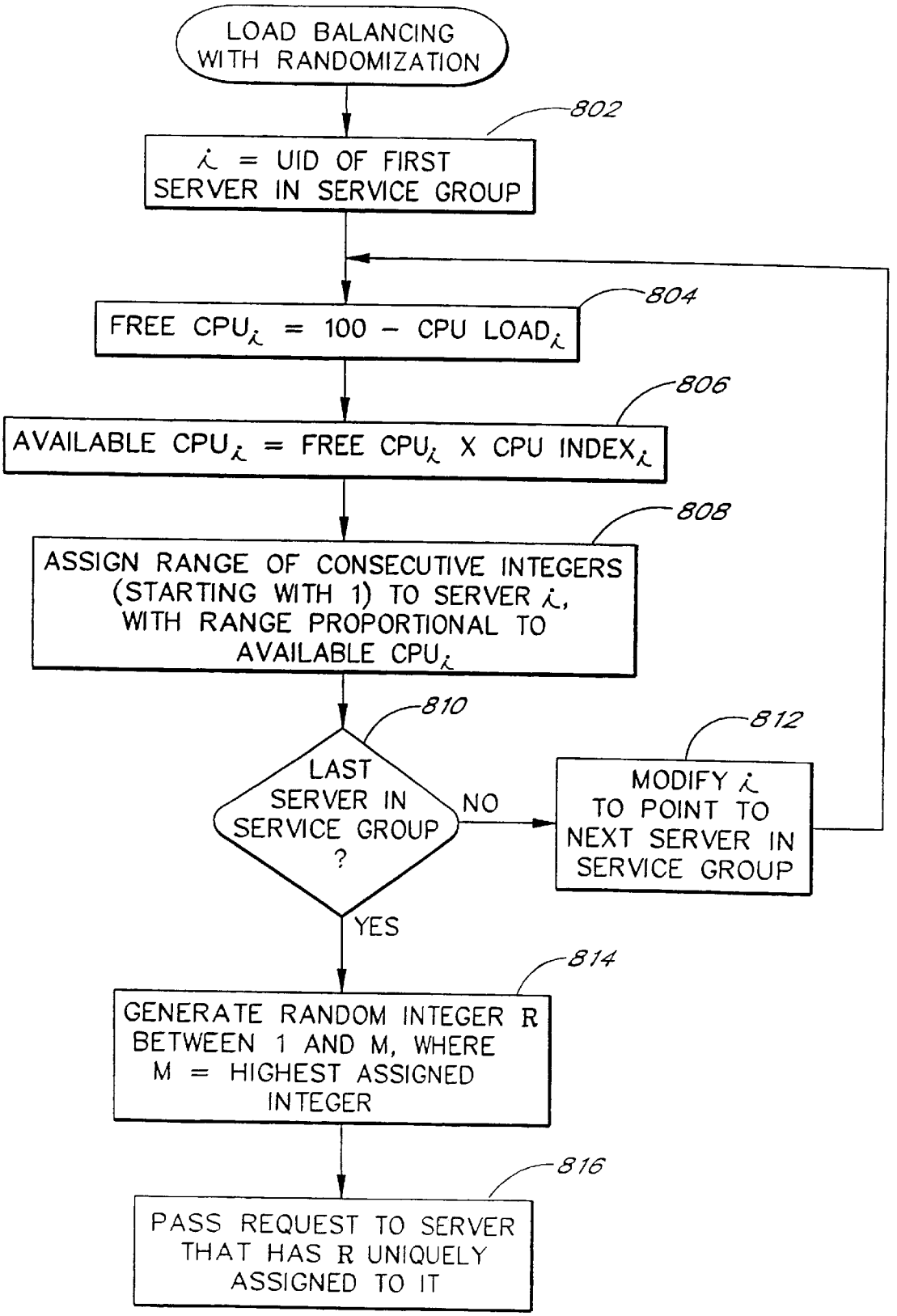

LOAD BALANCING WITH RANDOMIZATION
802
i = UID OF FIRST SERVER IN SERVICE GROUP
804
FREE CPU$_i$ = 100 − CPU LOAD$_i$
806
AVAILABLE CPU$_i$ = FREE CPU$_i$ X CPU INDEX$_i$
808
ASSIGN RANGE OF CONSECUTIVE INTEGERS (STARTING WITH 1) TO SERVER i, WITH RANGE PROPORTIONAL TO AVAILABLE CPU$_i$
810
LAST SERVER IN SERVICE GROUP ?
NO
812
MODIFY i TO POINT TO NEXT SERVER IN SERVICE GROUP
YES
814
GENERATE RANDOM INTEGER R BETWEEN 1 AND M, WHERE M = HIGHEST ASSIGNED INTEGER
816
PASS REQUEST TO SERVER THAT HAS R UNIQUELY ASSIGNED TO IT

FIG. 8

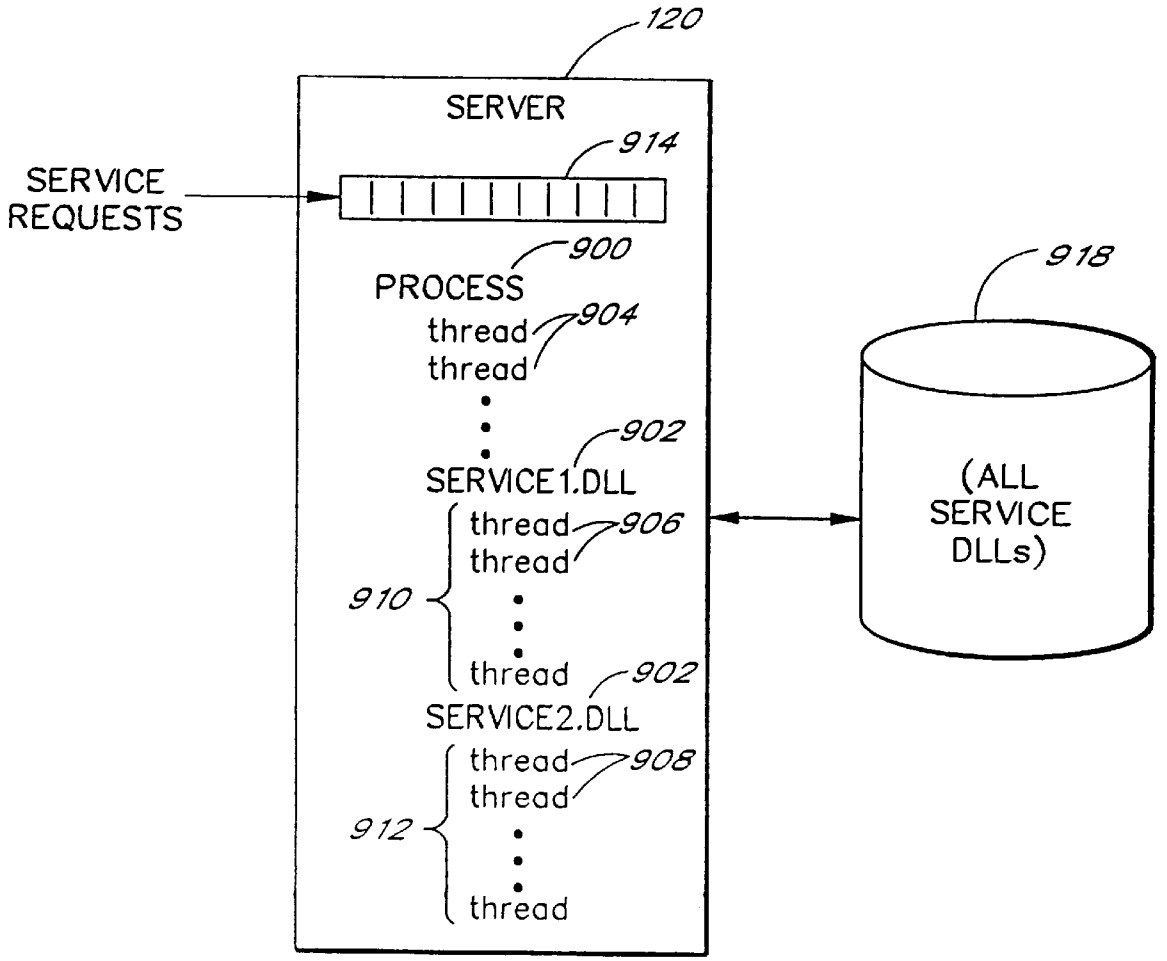

120
SERVER
914
SERVICE REQUESTS
900
PROCESS
thread
thread
904
902
SERVICE1.DLL
thread
thread
906
910
thread
902
SERVICE2.DLL
thread
thread
908
912
thread
918
(ALL SERVICE DLLs)

FIG. 9

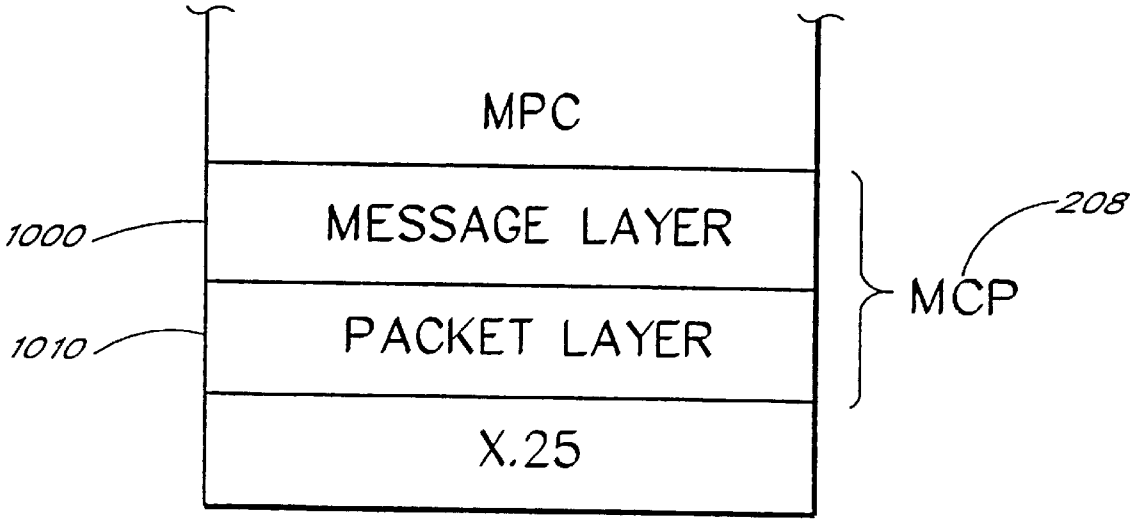


FIG. 10

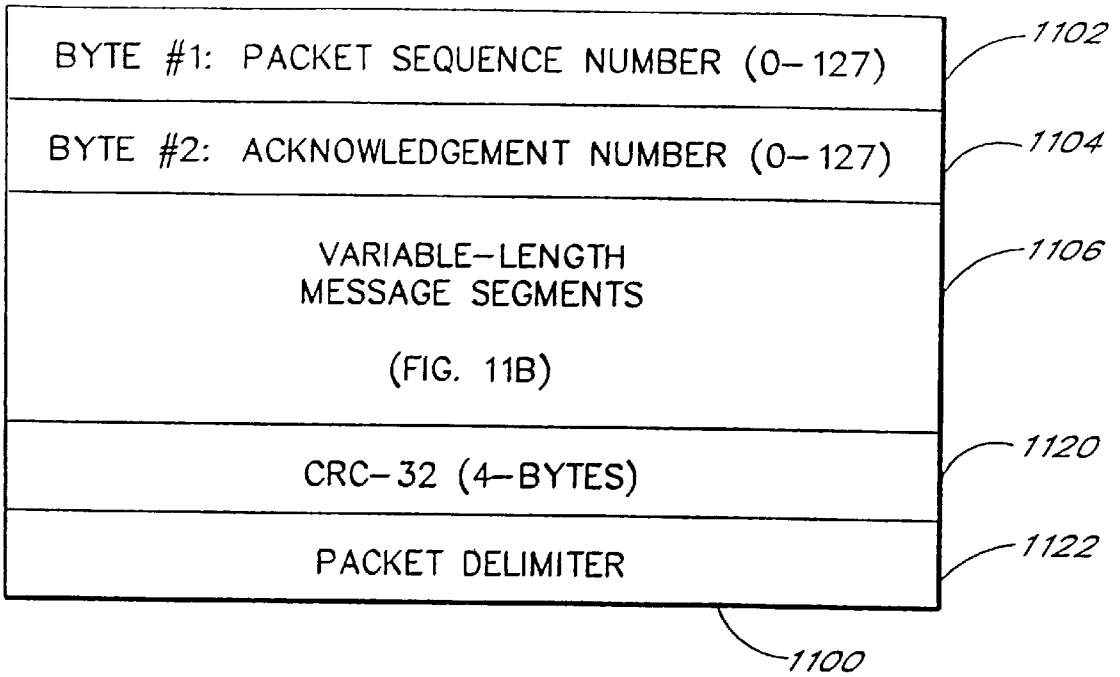


FIG. 11A

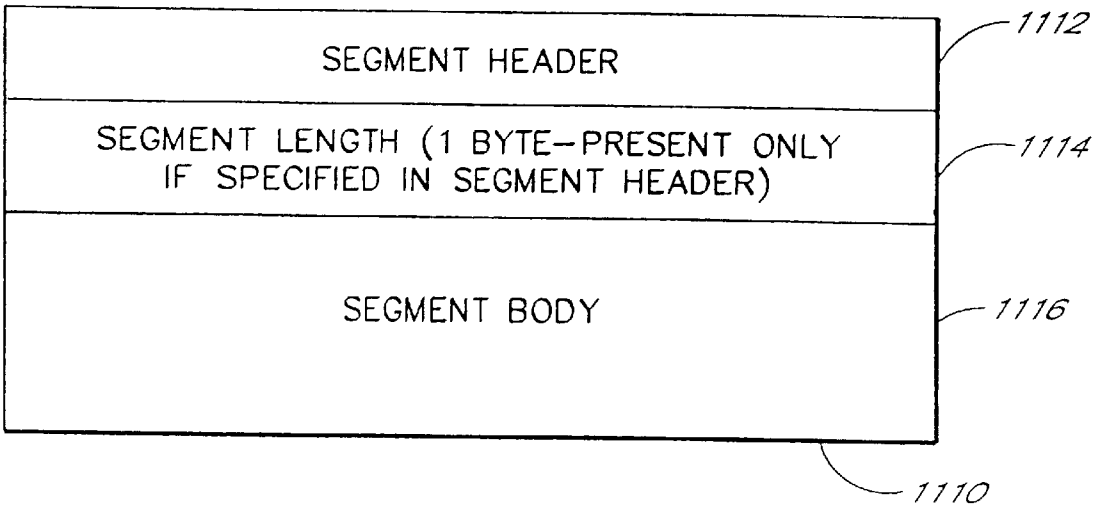


FIG. 11B

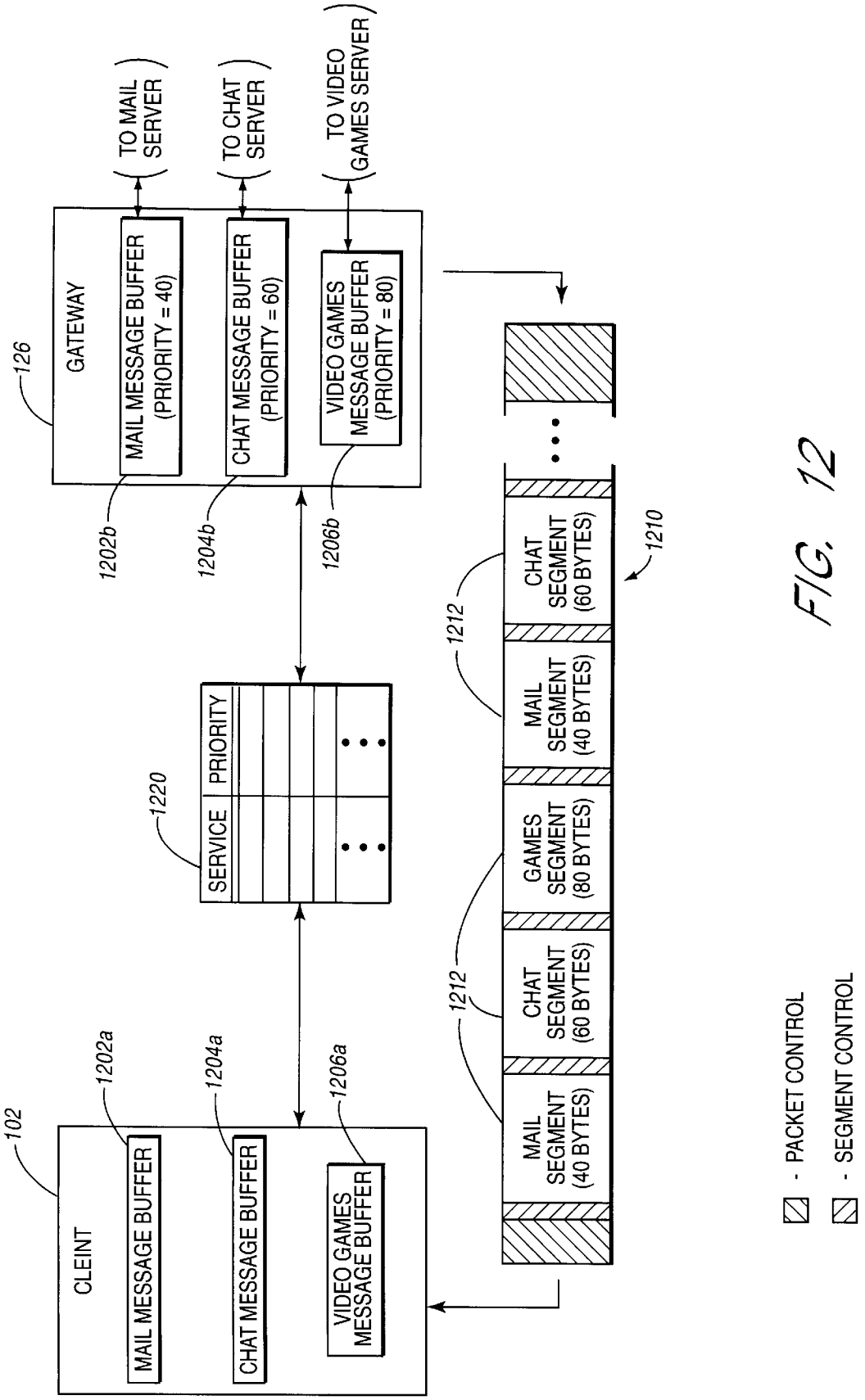

TO MAIL SERVER
TO CHAT SERVER
TO VIDEO GAMES SERVER
GATEWAY
126
MAIL MESSAGE BUFFER (PRIORITY = 40)
CHAT MESSAGE BUFFER (PRIORITY = 60)
VIDEO GAMES MESSAGE BUFFER (PRIORITY = 80)
1202b
1204b
1206b
SERVICE
PRIORITY
1220
CLEINT
102
MAIL MESSAGE BUFFER
CHAT MESSAGE BUFFER
VIDEO GAMES MESSAGE BUFFER
1202a
1204a
1206a
MAIL SEGMENT (40 BYTES)
CHAT SEGMENT (60 BYTES)
GAMES SEGMENT (80 BYTES)
MAIL SEGMENT (40 BYTES)
CHAT SEGMENT (60 BYTES)
1212
1210
- PACKET CONTROL
- SEGMENT CONTROL

FIG. 12

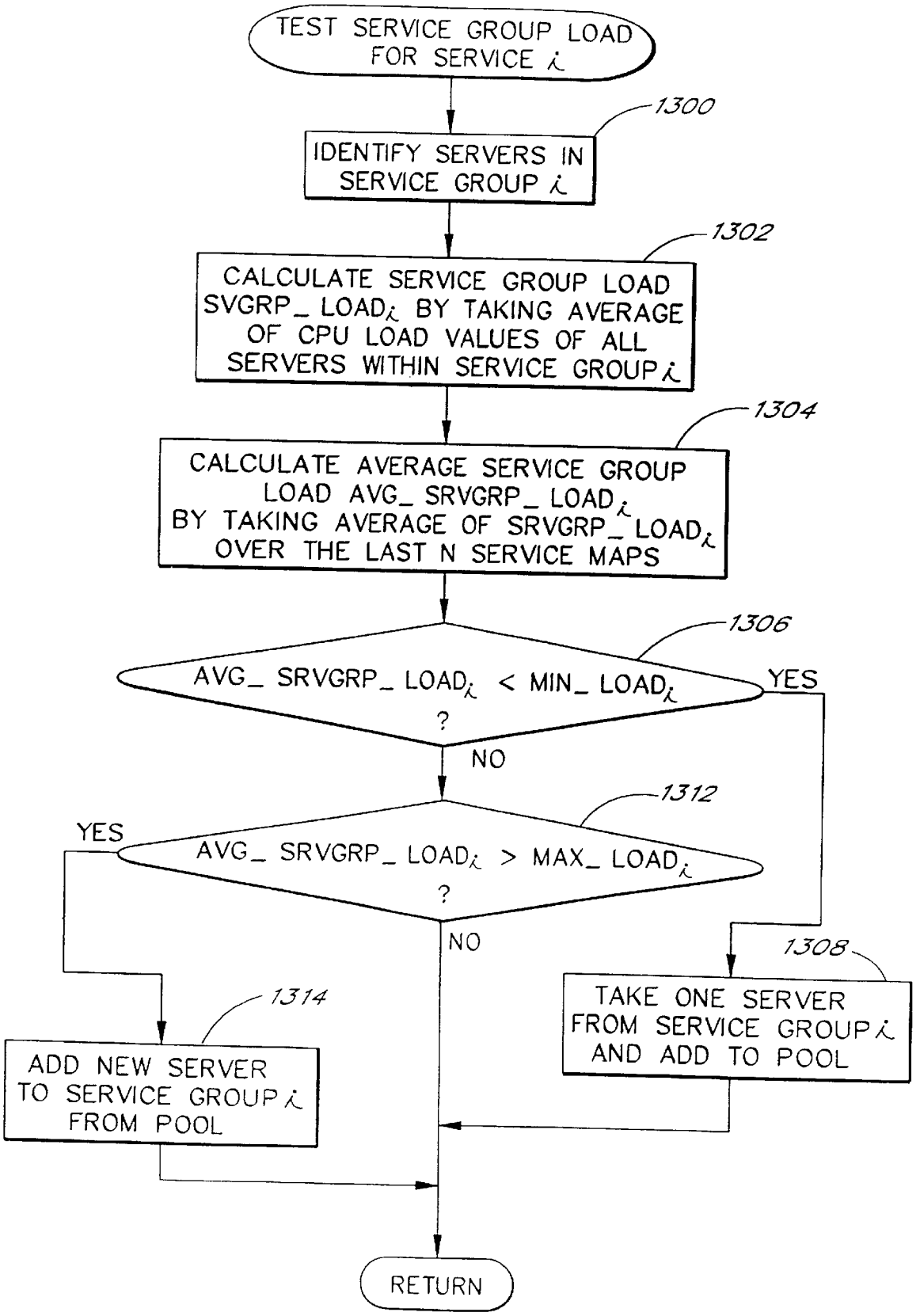

TEST SERVICE GROUP LOAD FOR SERVICE i
1300
IDENTIFY SERVERS IN SERVICE GROUP i
1302
CALCULATE SERVICE GROUP LOAD SVGRP_ LOADi BY TAKING AVERAGE OF CPU LOAD VALUES OF ALL SERVERS WITHIN SERVICE GROUP i
1304
CALCULATE AVERAGE SERVICE GROUP LOAD AVG_ SRVGRP_ LOADi BY TAKING AVERAGE OF SRVGRP_ LOADi OVER THE LAST N SERVICE MAPS
1306
AVG_ SRVGRP_ LOADi < MIN_ LOADi ?
YES
NO
1312
AVG_ SRVGRP_ LOADi > MAX_ LOADi ?
YES
NO
1308
TAKE ONE SERVER FROM SERVICE GROUP i AND ADD TO POOL
1314
ADD NEW SERVER TO SERVICE GROUP i FROM POOL
RETURN

FIG. 13

SYSTEM FOR ON-LINE SERVICE IN WHICH GATEWAY COMPUTER USES SERVICE MAP WHICH INCLUDES LOADING CONDITION OF SERVERS BROADCASTED BY APPLICATION SERVERS FOR LOAD BALANCING

FIELD OF THE INVENTION

The present invention relates to computer networks. More particularly, the present invention relates to large-scale computer networks for providing information-related services and communications services to end users.

BACKGROUND

Various on-line services networks are currently in existence, including Compuserve, Prodigy, and America On-Line. End users typically access these networks using a microcomputer equipped with a modem. During a logon session, a user can use a variety of information-related services and communications services, including news services, weather services, bulletin board services, email, and the like.

Existing architectures for on-line services networks suffer from a number of limitations. For example, existing on-line services networks do not provide an adequate means for allocating processing resources to user service requests, and thus do not make efficient use of available processing resources. Additionally, many existing on-line services networks do not provide a means for updating service applications without temporarily taking services off-line. Further, existing on-line services networks are not capable of allowing a user to access multiple services at a time.

It is an object of the invention to provide an on-line services network architecture which overcomes these and other limitations of the prior art.

SUMMARY OF THE PREFERRED EMBODIMENTS

In accordance with the present invention a client-server architecture for an on-line services network is provided in which on-line services are distributed and replicated on groups of application servers. The on-line services are implemented as client-server applications, with server portions (or "server applications") running on the application servers, and with client portions (or "client applications") running on client microcomputers of end users. The architecture features a high degree of scalability, allowing the capacity of the network to be scaled (without degradation of on-line services) as the number of end users increases over time. The architecture also features various mechanisms for dynamically balancing the processing load among the application servers of the system. The architecture also features various mechanisms for dynamically allocating processing resources (such as application servers) to specific on-line services, so that fluctuations in usage levels of specific on-line services can be efficiently accommodated. The architecture additionally features a transport mechanism that allows end users to simultaneously access multiple on-line services. The architecture further permits service applications to be updated without temporarily taking the services off-line.

In a preferred embodiment, the application servers of the system are interconnected by a local area network, and are arranged into service groups, with each service group corresponding to a particular service. Each application server of

a service group is preferably a "replicated" version of the other application servers within the service group, meaning that each runs the same server application (or server applications) as the others to implement a common service. For example, application servers within a bulletin board system ("BBS") service group all run a BBS server application. Depending upon the particular service, each application server of the service group may locally store a replicated copy of the service's content. For example, all application servers within the BBS service group preferably store identical copies of all BBS messages. A service group may have as few as one application server. Further, a given server can handle multiple services, and can thus be simultaneously allocated to multiple service groups.

Users of a service are assigned to individual application servers on a per-service session basis. For example, when an end user initially opens the BBS service, the user's BBS service session will be assigned to one of the replicated application servers of the BBS service group, and that application server will normally service the end user until the BBS service session is terminated.

Multiple Gateway microcomputers are also connected to the local area network. Each Gateway microcomputer is connected to a wide area network for communicating with client microcomputers of end users. The wide area network preferably uses CCITT X.25, and may additionally or alternatively use other standards such as TCP/IP, Frame Relay, or Asynchronous Transfer Mode (ATM). The Gateway microcomputers serve as intermediaries between the client microcomputers and the application servers. When a user initially logs onto the network, a connection is established between the client microcomputer and a Gateway microcomputer, and the connection is maintained throughout the logon session. During the logon session, the user typically opens and closes multiple services, and thus communicates with multiple application servers. The Gateway microcomputer assigned to the logon session maintains the various client-server message streams by routing service requests (from the client microcomputer) to the appropriate application servers, and by returning application server responses to the client microcomputer. In the preferred embodiment, each Gateway microcomputer can handle approximately 1000 simultaneous logon sessions.

In accordance with a dynamic load balancing feature of the invention, when a user sends a request to open a service, the Gateway microcomputer that receives the request initially identifies the application servers that are within the relevant service group. The Gateway microcomputer then determines the current load of each application server in the service group, and applies a load balancing method to select an application server that is relatively lightly loaded. The service request is then passed to the selected application server for processing.

In the preferred embodiment, the Gateway microcomputers identify the application servers in the service groups and determine the loads of the application servers by accessing a locally-stored service map. The service map contains information about every application server of the system. For example, for each application server, the service map indicates the names of the server applications running on the application server, the current processing load of the application server, and the overall processing power (in the form of a static CPU INDEX benchmark) of the application server. The service map is updated on the order of every 30 seconds to reflect changes in application server loads, and to reflect various changes in the system configuration.

Load balancing is preferably performed according to the specific dynamics of the individual services. One load

balancing method uses the load and CPU INDEX values contained in the service map to calculate the available CPU processing power for each application server in the relevant service group, and then passes the service request to the application server having the greatest available CPU processing power. Another load balancing method selects application servers such that the probability of selection of a given application server is proportional to that application server's available CPU processing power.

In accordance with another feature of the invention, a transport protocol that handles client-Gateway communications over the local area network multiplexes client-server message streams to permit end users to access multiple services simultaneously. Using this feature, a user can, for example, access one service while downloading a large file from another service. Multiplexing is preferably performed by combining variable-length segments of different message streams within wide area network packets, wherein each message stream corresponds to a different service. To accommodate for differing throughput requirements of different services, lengths of the variable-length segments are varied on a service-by-service basis according to predetermined (or user specified) priority levels. Different amounts of wide area network bandwidth are thereby allocated to different services.

In accordance with another feature of the invention, application servers may be dynamically allocated to services (or equivalently, to service groups) to accommodate for fluctuations in usage levels of different services. To allocate an application server to a particular service, the server application (or applications) that implement the service are dynamically loaded (i.e., loaded without stopping the underlying process) on the application server. Likewise, to remove an application server from a service group, the relevant server applications are dynamically unloaded. In the preferred embodiment, dynamic link libraries for every server application reside on the hard disks of every application server, so that any service can be activated on any application server at virtually any time. Allocation of application servers to service groups can be performed semi-automatically by a system operator who monitors service loads from a system console, or automatically using a load monitoring program that compares service loads to predetermined thresholds.

In accordance with yet another feature of the invention, a user service session can be transferred from one application server to another application server (which runs the same service application), without termination or interruption of the service session. Application servers can thus be taken off line (for maintenance, or for reallocation to another service group, for example) without interfering with the end user's immediate use of the on-line service.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features of the invention will now be described with reference to the drawings of certain preferred embodiments, which are intended to illustrate and not to limit the invention, and in which:

FIG. **1** is a high level diagram illustrating the architecture of an on-line services network in accordance with one embodiment of the invention;

FIG. **2** illustrates a client microcomputer concurrently communicating with two different application servers in accordance with a multiplexing feature of the present invention;

FIG. **3** illustrates, in example form, how different application servers can be assigned to different service groups;

FIG. **4** illustrates how a service map may be generated from server-specific local maps;

FIG. **5**A illustrates the primary software and hardware communications components used when a user connects to the network by modem;

FIG. **5**B illustrates the primary software and hardware communications components used when a user connects to the network using TCP/IP;

FIG. **6** is a flow chart illustrating the steps taken by a Gateway microcomputer upon receiving a request from a user to open a service;

FIG. **7** is a flow chart illustrating a load balancing technique used to select an application server to handle a request to open a service;

FIG. **8** is a flow chart illustrating a variation of the load balancing technique of FIG. **7**;

FIG. **9** illustrates a preferred software implementation of services on an application server;

FIG. **10** illustrates the two component layers of a protocol layer that is used for communications between client and Gateway microcomputers;

FIGS. **11**A and **11**B illustrate a preferred packet format used for data communications between client microcomputers and Gateway microcomputers over a wide area network;

FIG. **12** illustrates a multiplexing technique for combining multiple client-server message streams within packets transmitted over the wide area network; and

FIG. **13** is a flow chart illustrating a technique for monitoring service group loads.

Reference numbers in the drawings have three or more digits. The two least significant digits are reference numbers within the drawing, and the more significant digits are the figure number. For example, reference number **306** refers to an item first shown in FIG. **3**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Described herein is a distributed, client-server architecture which employs replicated application servers to provide services to end users. Although the architecture is described in the context of an on-line services network, it will be recognized that various aspects and features of the architecture are applicable to other types of distributed systems.

For convenience, the description is broken up into the following ten sections: ARCHITECTURAL OVERVIEW; SERVICE GROUPS; SERVICE MAP; COMMUNICATIONS COMPONENTS; DYNAMIC LOAD BALANCING BY ROUTING OF SERVICE REQUESTS; SOFTWARE IMPLEMENTATION OF SERVICES ON SERVERS; CLIENT-GATEWAY COMMUNICATIONS; ARBITER; DYNAMIC ALLOCATION OF SERVERS TO SERVICE GROUPS; and SCALABILITY.

1. Architectural Overview

FIG. **1** is a high level diagram illustrating the basic components of an on-line services network **100** in accordance with one embodiment of the invention. Multiple client microcomputers **102** are connected to a host data center **104** by a wide area network (WAN) **106**. The wide area network **106** includes WAN lines **108** which are provided by one or more telecommunications providers, and which allow end users (i.e., users the microcomputers **102**) over a wide geographic area to access the host data center **104** via modem. The WAN lines **108** preferably include both X.25 lines and ISDN (Integrated Service Digital Network) lines.

The X.25 lines **108** comply with the X.25 specification of the Comité Consultatif Internaionale de Télégraphie et Télé-

phonie (CCITT)—an international standards organization in Geneva, Switzerland that recommends communications standards. The X.25 standard documents the interface required to connect a computer to a packet-switched network. In addition to or in place of X.25, the wide area network **106** may use other types of communications standards, such as Frame Relay, or the Transport Control Protocol/Internet Protocol ("TCP/IP"—a suite of protocols developed for use on the Internet).

The host data center **104** comprises a plurality of application servers (APP servers) **120** connected to a high speed local area network (LAN) **122** (which may include multiple LANs, as described below). Each application server **120** has a unique server ID. Also connected to the LAN **122** are multiple Gateway microcomputers **126** (hereinafter "Gateways"), which link incoming calls from end users to the application servers **120**. In the preferred embodiment, the application servers **120** and the Gateways **126** are Pentium-class (or better) microcomputers which are scalable to at least four central processing units (CPUs), and which run the Microsoft Windows NT operating system available from Microsoft Corporation. The application servers **120** will typically have at least 128 MB of random-access memory (RAM) and at least 4 GB of disk space. Processing power may vary from application server to application server. For example, one application server may have four 120 MHz microprocessors, while another application server may have one 90 MHz microprocessor. Each Gateway **126** will typically have at least 64 MB of RAM and at least 2 GB of disk space, and will be capable of supporting approximately 1000 simultaneous user connections.

It is envisioned that the host data center **104** may advantageously have on the order of one hundred Gateways **126**, and between several hundred to several thousand application servers **120**. A host data center of this type will be able to handle tens of thousands of simultaneous user logon sessions. Advantageously, the processing capacity of the host data center **104** can easily be increased (to support increases in the number of subscribers, for example) by connecting additional Gateways **126** and application servers **120** to the LAN **122**, adding additional LANs when necessary. Further, additional host data centers **104** can be provided at different geographical locations to accommodate a wide geographic distribution of subscribers. The multiple host data centers **104** would preferably be interconnected by leased lines and/or by a data network (such as the Internet), and would preferably use TCP/IP to communicate.

The local area network **122** preferably comprises at least one 100 Mbps LAN which is based on the ISO/ANSI Copper Distributed Data Interface (CDDI) specification. (The CDDI specification is a variant of the well-known ANSI Fiber Distributed Data Interface specification, but uses a single copper ring instead of a dual fiber ring.) Various other types of LAN specifications may be used, such as IEEE token ring or switched ethernet. Further, various alternative transmission mediums may be used, including fiber optic lines.

To increase LAN bandwidth, some or all of the microcomputers of the host data center **104** may be interconnected by one or more additional LANs. For example, different LANs may be dedicated to different sets of application servers **120** (and to different services). Accordingly, the term "local area network" (or "LAN"), as used herein, should be construed to include a combination of local area networks, and should not be construed to be limited to a particular data transfer protocol or transmission medium.

The host data center **104** provides a variety of communications-based and information-based on-line services to end users. Typical services include, for example, a mail service for allowing users to send email messages to one another, a bulletin board system (BBS) service for allowing users to post and review messages on specific topics, a chat service for allowing users to communicate in real time with each other on specific topics, an interactive games service for allowing users to compete against each other in real time in on-line interactive games, various news services for allowing users to access news and magazine articles, a mediaview service for allowing users to view on-line multimedia titles, and a directory service for allowing users to view a hierarchy of services and data entities on the network. As described below, the server side of each on-line service is preferably implemented using one of the following: (1) a single application server **120**, (2) a set of "replicated" application servers (i.e., application servers which run the same service application or applications) that provide access to replicated (and locally-stored) copies of service "content" data (i.e., data provided to end user's of the service), or (3) a set of replicated application servers that provide access to server-specific (non-replicated) service content data.

The host data center **104** also includes multiple Arbiter microcomputers **128** that monitor, record and process certain types of transactions to ensure consistency among replicated application servers. The various functions performed by the Arbiter microcomputers **128** are described below under the headings SERVICE GROUPS and ARBITER. The host data center **104** also includes one or more custom Gateway microcomputers **130** which link the host data center **104** to one or more external service providers **132**, such as a credit card service that validates and executes credit card transactions. Each custom Gateway microcomputer **130** uses the communications protocol required by the external service provider **132** to which the custom Gateway is linked. In other embodiments, the functionality of the custom Gateways **130** may be integrated into the application servers **120**.

The host data center **104** also includes a number of administrative servers **134**. The administrative servers **134** perform administrative functions such as accounting, billing, network management, backup, system security, performance analysis, and server-to-service allocation. Because the description that follows focuses primarily on the application servers **120** (rather than the administrative servers **134**), the term "server" will hereinafter be used to refer to an application server **120** (unless indicated otherwise).

The on-line services offered to end-users of the network **100** are in the form of client-server applications programs (or "service applications"). Each service application includes a server portion (also referred to as a "server application") that runs on one or more of the servers **120**, and a client portion (also referred to as a "client application") that runs on a microcomputer **102** of an end user.

In the presently preferred embodiment, the client applications are in the form of Microsoft Windows '95 executables. These client applications are preferably written such that, from the end user's perspective, the various on-line services provided by the host data center **104** appear transparently as an extension to the file structure of the user's hard disk. Server applications are preferably in the form of service dynamic link libraries (DLLs) that are executed by the application servers **120**. Although the term "service application" will be used herein to describe applications running on servers **120**, it should be understood that only a server portion of the service application is actually running on the servers **120**, and that the service application

also includes a client portion running on client microcomputers **102** (or running on other computers of the host data center **104** that act as clients).

In accordance with one feature of the invention, service DLLs can be dynamically loaded and unloaded from the servers **120**, thereby allowing different services to be efficiently (and automatically) activated and deactivated on different servers. The preferred process by which services are activated and deactivated on different servers is described below under the headings SOFTWARE IMPLEMENTATION OF SERVICES ON SERVERS and DYNAMIC ALLOCATION OF SERVERS TO SERVICE GROUPS.

The term "service" refers generally to the abstract function or functions performed by a particular service application (or group of related service applications). Services generally fall within two categories: "on-line" services that are "seen" by end users when running client applications on client microcomputers **102**, and "background" services which are provided to other components of the host data center **104** (when, for example, one server **120** performs a function for another server **120**.) Although much of the following description focusses on on-line services, it should be recognized that many features of the invention are applicable to both on-line services and background services.

The architecture of the present invention is characterized as having different servers **120** running different service applications. The various on-line services (and background services) are thus distributed among the servers **120** of the host data center **104**. When a server runs a particular service application (or group of related service applications), it is said to be "allocated" to the corresponding service.

In accordance with one feature of the invention, service applications are replicated across multiple servers to form "service groups." The servers of a service group run the same service application (or set of related service applications), and therefore implement the same service (or set of related services). Although multiple servers **120** may be allocated to the same service (or equivalently, to the same service group), a user of the service will normally be serviced by only one of the servers **120** of the service group.

In accordance with another feature of the invention, when a particular service becomes heavily loaded, additional servers are allocated to the heavily-loaded service (by adding servers to the service group) to assist in handling the increased load. As described below under the heading DYNAMIC ALLOCATION OF SERVERS TO SERVICE GROUPS, allocations of servers to services can be performed either by a system operator who monitors the service loads from a control station, or by load monitoring software which compares service loads to predetermined thresholds.

To facilitate the operational description that follows, the following terminology and conventions will be used. The term "client-user" will be used to refer to a client microcomputer **102** under the control of an end user of the on-line services network **100**. Names of specific on-line services (and on-line service applications) will be written in capital letters (for example, "CHAT" or "MAIL"), and names of background services will be written in mixed case (for example, "Arbiter" or "Conf_Loc"). Further, the convention {service name}.DLL (for example, "CHAT.DLL") will be used to refer to specific dynamic link libraries used to implement the server portions of service applications.

During a typical logon session, a client-user will maintain a communications link with a single Gateway **126**, but may access multiple services (and thus communicate with multiple servers **120**). To initially access a service, the client-user sends an "open" request to the service. To terminate a service, the client-user sends a "close" request to the service. The use of a particular service by a single client-user (from the opening of the service to the closing of the service) will be referred to herein as a "service session."

Each time the user opens a service, the Gateway **126** that is handling the logon session accesses a locally-stored service map **136** to select a server **120** that is allocated to the particular service, and then establishes a service instance channel with the selected server **120**. The service instance channel is preferably in the form of a TCP/IP link, and is recorded in a session map **522** (shown in FIGS. **5**A and **5**B) on the Gateway **126**.

The service instance channel is maintained throughout the service session, and is used to pass messages between the Gateway **126** and the server **120** as the client and server portions of the service application interact. When multiple simultaneous service sessions are established between a given Gateway **126** and a given server **120**, the corresponding service instance channels are multiplexed over a single TCP/IP channel. Messages sent from the client microcomputer **102** to the server **120** are normally in the form of service requests (i.e., requests to have particular services performed by the server portion of the service application). Messages sent from the server **120** to the client microcomputer **102** are normally in the form of responses to specific service requests.

During a service session, the Gateway **126** performs two basic functions. First, the Gateway **126** performs protocol translation, translating messages between the protocol of the WAN **106** (such as X.25) and the protocol of the LAN **122**. Second, the Gateway routes messages between WAN channels and LAN channels so as to maintain communications between the client microcomputer **102** and the selected server **120**.

Advantageously, the preferred transport layer (referred to herein as the Microsoft Connection Protocol layer, or "MCP layer") that manages client-Gateway communications supports simultaneous service sessions, allowing a client-user to access multiple services simultaneously. Thus, for example, a user can access one on-line service while downloading a large software update file from another on-line service. This feature of the present invention is illustrated in FIG. **2** for a user who has opened two services, CHAT and WEATHER. The CHAT and WEATHER client applications **200** and **204** generate message streams that are passed to an MCP layer **208***a* that is running on the client microcomputer **102**. The MCP layer **208***a* multiplexes these two message streams and sends the multiplexed data to a Gateway **126** at the host data center **104**. The MCP layer **208***b* running on the Gateway **126** demultiplexes the data, and routes the respective message streams (over the LAN **122**) to the appropriate servers **120***a* and **120***b*. Likewise, the MCP layer **208***b* on the Gateway **126** multiplexes the message streams that are generated by the CHAT and WEATHER servers **120***a* and **120***b*, and the MCP layer **208***a* on the client microcomputer **102** demultiplexes these message streams and passes them to the respective client applications **200** and **204**. In the example shown, the user could, for example, download a large bitmap image of a weather satellite photo while simultaneously communicating in real time with another CHAT user. The MCP layer is further described below under the heading CLIENT-GATEWAY COMMUNICATIONS.

2. Service Groups

As indicated above, when multiple servers **120** are allocated to the same service they form what is referred to herein as a service group. The creation of a service group is

desirable, for example, when the processing load placed on a service cannot be adequately handled by a single server **120**. Normally, all servers **120** within a service group are replicated, meaning that they all run the same service application (or set of service applications) that implements the server portion of a particular service (or set of related services). Each client service session can thus be handled by a single server **120** within the service group.

For certain types of on-line services, all replicated servers of the service group maintain local copies of the service's content data, and provide user access to such data. Because all such servers run like service applications and provide access to like service data, the service appears the same to all users, even though different user service sessions may be assigned (by the Gateways **126**) to different servers **120** of the service group.

It should be recognized from the foregoing that although a service session will normally be handled by a single server **120**, the entity that provides the service as a whole is the service group. The service performed by a service group can be either an on-line service (such as CHAT or MAIL), or a background service (such as the Conf_Loc service which is used to locate CHAT conferences, as described below).

An example allocation of servers **120** to services is illustrated in FIG. **3**. In this example, four application servers **120***a–d* have been allocated solely to a MAIL service and two application servers **120***f* and **120***g* have been allocated solely to a BBS service. Server **120***e* is allocated to both the MAIL service and the BBS service, and thus runs both MAIL and BBS service applications. In this example configuration, servers **120***a–e* fall within the MAIL service group **300**, and servers **120***e–g* fall within the BBS service group **304**. Further, servers **120***a–e* are all replicated of one another (meaning, in this example, that they all run the same MAIL service application), and servers **120***e–g* are all replicated of one another (meaning that they all run the same BBS service application).

Certain types of on-line services provide end users with access to service-wide (or equivalently, service-group-wide) data sets (also referred to generally as "service content data") which change on a transaction-by-transaction basis. The BBS service is an example of such a service; each time a user posts a message, a data set is modified such that all users of the BBS service (regardless of the particular BBS server to which a user is assigned) have access to the message. Transactions of this general type (i.e., transactions which affect service-wide data sets) will hereinafter be referred to as "update transactions," and can originate from a variety of sources. For example, an update transaction can originate from a client microcomputer **102** when an end user posts a BBS message, or can originate from an external data provider that is connected to the host data center **104** by the Internet.

Although service content data could be stored in a central data base which is accessible to all servers **120** of the corresponding service group, for performance reasons, it is normally preferable (depending upon the particular service) to have each server **120** of the service group locally store redundant copies some or all of the service's content data. To ensure consistency between such locally-stored data sets (so that all users of the service see the same data), an Arbiter service is used to replicate update transactions across all servers **120** of a service group. In the preferred embodiment, the Arbiter service runs on the Arbiter microcomputers **128**, and one Arbiter microcomputer **128** is allocated to each service (i.e., to each service group) that uses the Arbiter service to replicate update transactions.

In the preferred implementation of the BBS service, the Arbiter service is used to replicate message postings. Thus, in the example configuration shown in FIG. **3**, a BBS Arbiter microcomputer **128** is allocated to the BBS service group **304**. To accommodate increased numbers of BBS users, multiple BBS service groups (not shown) may be formed, with each BBS service group handling a specific topic or set of topics. A separate Arbiter microcomputer **128** would then preferably be used for each BBS service group.

In the preferred implementation of the MAIL service, each user is assigned to a particular MAIL server **120** (i.e., the user's mailbox is stored on only one MAIL server), and email messages directed to a user are forwarded to the user's MAIL server by the other MAIL servers. Thus, the replication services of the Arbiter are not required, and no Arbiter microcomputer is allocated to the MAIL service group **300**.

The Arbiter service is further described below under the heading ARBITER.

3. Service Map

Referring again to FIG. **1**, in order to route client-user service requests to the appropriate servers **120**, the Gateways **126** must have some way of determining the unique IDs of the servers that are currently handling the requested services. This is accomplished by means of the service map **136**, which contains information about every service and server **120** in the host data center **104**. The service map **136** also contains information about the current load of each server **120** in the host data center **104**. This load information is used to uniformly distribute the service load among the servers **120**, as further described below under the headings DYNAMIC LOAD BALANCING BY ROUTING OF SERVICE REQUESTS and DYNAMIC ALLOCATION OF SERVERS TO SERVICE GROUPS.

Still referring to FIG. **1**, the service map **136** is preferably generated by a service map dispatcher **144**, which may be implemented on a single microcomputer. To generate the service map **136**, each server **120** periodically generates a local map **140**, and transmits the local map **140** to the service map dispatcher **144**. Each local map **140** contains up-to-date information about the respective server **120** that generated the local map, including information about the server's current load. The service map dispatcher **144** builds the service map **136** from all of the local maps **140** it receives, and then broadcasts the service map **136** to all of the Gateways **136** over the LAN **122**. In other embodiments, the servers **120** broadcast their respective local maps **140** to the Gateways **126**, and each Gateway builds the service map **136** from the local maps it receives.

In the preferred embodiment, the service map dispatcher **144** broadcasts a new service map **136** every **30** seconds. Each time a new service map **136** is broadcasted, every Gateway **126** receives and locally stores a copy of the new service map **136**, overwriting old versions of the service map. The Arbiters microcomputers **128** also receive and store copies of the service map **136**, and use the service map to determine the states of the servers **120** that are currently assigned to the respective service groups.

FIG. **4** illustrates the information contained within each server-specific local map **140**, and also illustrates the process by which the service map dispatcher **144** builds the service map **136** from the local maps **140**. The service map **136** contains the local map **140** for each operational server **120** within the host data center **104**, along with the corresponding server IDs (indicated by the numbers **1**, **2**, . . . n) for the servers **120**.

Each local map **140** contains a CPU LOAD value and a CPU INDEX value for the respective server **120**. The CPU

LOAD indicates the current load of the server **120**. The CPU LOAD is a numerical value, preferably ranging from 0 to 100, which indicates the percentage of available time that the CPU (or CPUs) of the server **120** is processing service requests. A CPU LOAD of zero indicates that the server **120** is minimally loaded, and a CPU LOAD of 100 indicates that the server is heavily loaded. In the preferred embodiment, the CPU LOAD is periodically calculated by the operating system (NT) on each server **120**, and is taken as an average over a 5 second period. The CPU LOAD values contained in the local maps **140** are periodically updated to reflect changes in server loads.

The CPU INDEX is a static benchmark value, indicating the processing power of the particular server **120**. In the preferred embodiment, the CPU INDEX is a benchmark that measures the server's performance under Microsoft Windows NT, although other conventional benchmarks could be used. The CPU INDEX accommodates for servers **120** of varying processing powers, and can be omitted in system configurations for which all servers **120** have substantially the same processing power.

Together, the CPU LOAD and CPU INDEX for a server **120** indicate the available processing power for the server, and are thus useful for identifying the servers **120** that are most capable of handling new service sessions. The various methods for selecting servers **120** based on CPU LOAD and CPU INDEX values are described below under the heading DYNAMIC LOAD BALANCING BY ROUTING OF SERVICE REQUESTS.

With further reference to FIG. **4**, each local map **140** contains at least one service description **400**, and contains multiple service descriptions **400** (as shown in FIG. **4**) if the server **120** is currently allocated to multiple services. Each service description **400** includes a service name, the state of the service DLL (launching, active, active-accepting, or stopped), the client count (i.e., the number of client-users that are currently assigned to the server **120** to use the service), and the version number of the service DLL. The service descriptions **400** contained within the service map **136** are used for various purposes by different components of the host data center **104**. For example, the service descriptions **400** are used by the Gateways **126** to locate servers **120** for handling client service requests.

The service map **136** advantageously serves to inform the Gateways **126** (and other components of the on-line services network **100**) of the addition, deletion or change in state of any server **120** in the system. The use of the service map **136** thereby allows changes to be made to the servers **120** of the system without manual intervention. By way of example, assume that a new server **120** is added to the host data center **104**, and is initially configured to act as a MAIL server. Shortly after being brought up, this new MAIL server will generate and send a local map **140** to the service map dispatcher **144**. The service map dispatcher **144** will then broadcast the local map **140** (as part of the next service map **136**) to all of the Gateways **126**, allowing the Gateways to send service requests to the new MAIL server. Thus, once the new server **120** has been configured, its addition to the system is automatic, requiring no system operator intervention.

In addition to generating a service map **136**, the service map dispatcher **144** maintains a central repository of information referred to as the "global registry" **145**. The global registry **145** contains various information about the present configuration of the host data center **104**. For example, for each service group, the global registry **145** indicates the IDs of the servers **120** of the service group, and the identity of the Arbiter microcomputer **128** (if any) which is assigned to the service group. Of course, the global registry **145** could be maintained by some entity other than the service map dispatcher **144**.

4. Communications Components

FIGS. **5**A and **5**B illustrate the basic communications components, including the protocol layers provided within the client, Gateway, and server microcomputers **102**, **126**, **120**, for a preferred embodiment of the on-line-services network **100**. The communications components shown in FIG. **5**A are used when an end user logs onto the network via modem **510**. The communications components shown in FIG. **5**B are used when the end user logs onto the network via a TCP/IP connection (such as an ISDN link).

With reference to FIGS. **5**A and **5**B, client applications **500***a* make use of a high-level application programming interface (API), referred to herein as the Microsoft Procedure Call layer, to communicate with server portions of service applications **500***b*. The Microsoft Procedure Call layer (hereinafter "the MPC layer") is an optimized remote procedure call (RPC) layer, optimized to permit efficient client-server communications over a relatively slow WAN. Unlike conventional RPC, the MPC layer allows a client application to send out multiple requests without receiving a response, and allows each request to be individually monitored and/or canceled from the client side. The MPC layer is shown in FIGS. **5**A and **5**B as comprising respective layers **502***a* and **502***b* within the client microcomputer **102** and the server **120**. (Depending upon the context in which it is used herein, the term "layer" may refer either to a specific set of structures and routines on a particular machine, or may refer to the general functionality of such structures and routines as a whole.) Requests from the MPC layer **502***a* are passed to the MCP layer, which is shown in FIGS. **5**A and **5**B as comprising respective layers **208***a* and **208***b* in the client microcomputer **102** and the Gateway **126**. The MCP layer is seen by the MPC layer as providing reliable end-to-end client-Gateway communications. The MCP layer handles various aspects of client-Gateway communications, including packetization (for X.25 connections), segmentation and multiplexing of message streams, flow control (using a sliding window algorithm), and error correction (using CCITT CRC-32). In addition to supporting X.25, the MCP layer supports TCP/IP user connections (FIG. **5**B). In accordance with one feature of the invention, when a client-user concurrently accesses multiple services, the MCP layer allocates WAN bandwidth to services based on the differing throughput requirements of the services. In addition to implementing a preferred communications protocol, the MCP layer also includes a client API and a server API for interfacing respectively with client and server portions of service applications. The MCP layer is further described below under the heading CLIENT-GATEWAY COMMUNICATIONS.

With reference to FIG. **5**A, when the client-user accesses the on-line services network **100** by a modem **510**, the MCP layer **208***a* communicates with a modem engine layer **508**, which in-turn communicates with the modem **510**. The modem **510** communicates over a telephone line **512** with a carrier-provided PAD (packet assembler-disassembler) **514** that translates data between a standard modem format and the WAN format (which is X.25 in the FIG. **5**A example). At the Gateway **126**, an X.25 network card **516** sends and receives data over an X.25 line **108**, and communicates with an X.25-specific network layer **518**. With reference to FIG. **5**B, when the user accesses the on-line services network **100** via a TCP/IP connection, the MCP layer **208***a* on the client

microcomputer communicates with a network layer **508***a*, and the network layer **508***a* communicates over the WAN **106** with a TCP/IP card **516***a* of the Gateway **126**.

Within the Gateway **126**, a redirector layer **519** communicates with a locator program **520**. The redirector layer **519** and the locator program **520** perform the functions of (1) selecting servers **120** for handling requests to open services (referred to herein as "open" requests), and (2) routing the "open" requests so as to distribute processing loads within service groups. (As indicated above, a client-user will typically generate a number of "open" requests throughout a logon session, since the user will typically access multiple services. Additional service requests will normally be generated by the client-user throughout the course of each service session.) The locator program **520** accesses a locally-stored copy of the service map **136** whenever an "open" request is received from a client microcomputer **102**. Each time a server **120** is selected to handle an open request, the redirector layer **519** records the selected server's ID within a session map **522**. In the course of an ongoing service session, the Gateway **126** accesses its session map **522** whenever a client-server message is received, and uses the session map to properly route the message.

The redirector layer **519** communicates with the MCP layer **208***b* to communicate with client microcomputers **102**, and communicates with a Microsoft Host Connection Protocol (MHCP) layer **510***a* to communicate with servers **120**. The MHCP layers **510***a* and **510***b* perform various transport-layer-type functions, such as formatting of message data for transmission over the LAN **122**, and multiplexing of message data for transmission over TCP/IP links of the LAN **122**. The MHCP layers **510***a*, **510***b* communicate with CDDI network layers **524***a* and **524***b*, which in-turn communicate with CDDI network cards **530***a* and **530***b*.

It will be recognized that although the implementation depicted by FIGS. **5**A and **5**B use specific communications protocols and protocol layers, various alternative protocols and protocol layers could be used.

5. Dynamic Load Balancing by Routing of Service Requests

FIG. **6** illustrates a preferred sequence of steps that is performed whenever a Gateway **126** receives an "open" request from a client-user. As indicated by blocks **600** and **602**, the Gateway **126** initially accesses the current service map **136** to determine the number of servers **120** that are allocated to the requested service. If only one server **120** is currently assigned to the service, the Gateway routes the service request to that server, as indicated by blocks **604** and **620**. If the number of servers **120** assigned to the service is zero, an error code is returned, as indicated by block **620**.

If multiple servers **120** are assigned to the service group, the Gateway **126** then determines whether or not the particular service requires load balancing, as indicated by the decision block **606**. Although load balancing is appropriate for most services, certain services require that service requests be handled by a particular server **120**, and thus require the use of a custom locator program (as indicated by block **610**) to identify the correct server.

An example of a service which requires a custom locator program is CHAT. Although multiple servers **120** can be assigned to the CHAT service, each CHAT conference is preferably handled by a single server **120** (to facilitate communications between users who have joined the CHAT conference). Thus, when an end user generates a request to join a specific CHAT conference (by, for example, specifying a particular CHAT topic), the identify of the particular CHAT server **120** that is handling the CHAT conference must be determined. In the preferred embodiment, the CHAT server **120** is identified using a special Conf_Loc (conference locator) service, which is a background service (running on one or more of the servers **120**) that maps CHAT resources to servers. To identify the CHAT server **120**, the following step are performed:

(a) The client (i.e., the client application running on a user's microcomputer **102**) sends an "open" request to the Conf_Loc service. The Gateway **126** applies load balancing (in accordance with the flowchart of FIG. **6**) to select a Conf_Loc server **120** to handle this open request.

(b) The client sends a request to the Conf_Loc service (which is routed by the Gateway **126** to the Conf_Loc server identified in step a) to return the server ID corresponding to the identified CHAT conference, and waits until the requested server ID is returned. (If the requested CHAT conference does not exist, the Conf_Loc service will start a new CHAT conference and return the ID of the server that will be handling the new CHAT conference).

(c) The client closes the Conf_Loc service.

(d) The client sends an "open" request to a Conf_Srv service with a parameter specifying the CHAT server ID. The custom locator for the Conf_Srv service on the Gateway **126** parses this parameter and returns the CHAT server ID to the redirector layer **519** on the Gateway. This connects the client to the CHAT server.

In an alternative embodiment, the client sends an "open" request to the CHAT service with a parameter specifying the conference name, and A CHAT custom locator uses this parameter to identify the server ID of the CHAT server **120**, starting a new conference if necessary. The CHAT custom locator then returns the server ID of the CHAT server to the redirector layer **519**. This alternative method does not require the use of a separate Conf_Loc service.

Another example of a service which requires a custom locator program is MAIL. (As described above, each user's mailbox is preferably stored on only one of the MAIL servers.) When a user requests access to his or her mailbox, a MAIL locator service is provided with the username (or other user identifier) of the end-user, and the MAIL locator returns the server ID of the mail server which stores the user's mailbox.

With reference to blocks **612** and **614**, if the requested service does not require the use of a custom locator program, the Gateway **126** reads the service map **136** to obtain the CPU LOAD and CPU INDEX values for every server within the service group. The Gateway then accesses a table which specifies a load balancing method to use for the service. Although the same load balancing method could be used for all services, it has been found that a more uniform load balancing result can be achieved by taking into account the differing processing demands placed on the servers **120** by different service applications. Two load balancing methods, and some variations of these methods, are discussed below.

With reference to blocks **618** and **620**, once a load balancing method has been selected, the method is applied to select a server **120** within the service group. The "open" request is then passed to the selected server.

FIG. **7** illustrates a first preferred load balancing method, which selects the server **120** which most recently had the highest available CPU processing power. As illustrated by blocks **702–710**, the CPU LOAD and CPU INDEX values for each server **120** in the service group are used to calculate an AVAILABLE CPU value for each such server. The available CPU processing power (AVAILABLE CPU) is calculated for a server **120** by multiplying the server's CPU

INDEX by the server's FREE CPU, where FREE CPU is a value ranging from 0 to 100 which is simply calculated by subtracting CPU LOAD from 100. With reference to block **712**, once AVAILABLE CPU has been calculated for each server **120** in the service group, the server with the highest AVAILABLE CPU is selected to handle the service request.

The method illustrated in FIG. **7** works well provided that the actual server loads remain relatively static between service map updates (which occur every **30** seconds in the preferred embodiment), so that the CPU LOAD values in current service map can be relied upon. For certain types of services, however, relatively small changes in the number of ongoing service sessions can cause significant changes in server loads. An example of such a service might be an interactive video games service which requires a significant amount of server processing power per user. With the load balancing method of FIG. **7**, all requests for the service over a period of **30** seconds would be routed to the same server **120** within the service group, potentially overloading the server. This problem could be alleviated by updating the service map **136** more frequently, but at the cost of increased use of available bandwidth on the LAN **122**.

A preferred solution to this problem is to employ a randomization technique in selecting a server **120**. FIG. **8** illustrates a preferred load balancing method which uses randomization. As with the method of FIG. **7**, the Gateway **126** calculates the AVAILABLE CPU for each server **120** within the service group, as indicated by blocks **802–812**. However, as indicated by block **808**, rather than selecting the server **120** with the highest AVAILABLE CPU, each server is assigned a unique range of consecutive (positive) integers such that the number of integers assigned to a server is proportional to the server's AVAILABLE CPU. In assigning ranges, the first assigned range of numbers begins with 1, and each successive range begins with the lowest unassigned integer. By way of example, assume that the service group consists of three servers, Server A, Server B and Server C, and that Server A has twice the available CPU processing power of Server B and four times the available CPU processing power of Server C. According to the method, the ranges integers could be assigned as follows: Server A←1–200; Server B←201–300; Server C←301–350.

As indicated by block **814**, a pseudo-random integer R between 1 and M is then generated (using a standard pseudo-random number generator) by the Gateway **126**, where M is the highest assigned integer (350 in the above example) . As indicated by block **816**, the server **120** that has the integer R assigned to it is then selected to handle the "open" request. By selecting servers in this manner, the probability of selection of a server is proportional to the available CPU processing power of that server.

As will be recognized by those skilled in the art, various alternative load balancing methods are possible. It will further be recognized that the load balancing function need not be performed by the Gateways **126**. For example, the Gateways **126** could be programmed to pass each "open" request to one of the servers **120** within the service group, and the recipient server could apply the load balancing method and (if necessary) reroute the request.

Although the load balancing methods of FIGS. **7** and **8** use a specific type of load data to balance the processing load among the servers **120**, it will be recognized that various other types of load data could be used. For example, the current client count for each server **120** could be used as a basis for load balancing.

6. Software Implementation of Services on Servers

As indicated above, server portions of service applications are preferably in the form of service DLLs (dynamic link libraries) which run under the Microsoft Windows NT operating system. Typically, a server **120** will be assigned to only a single service at a time. However, as indicated above, one server **120** can run multiple service applications concurrently, and can thus be assigned to two or more services simultaneously. The ability for a server **120** to handle multiple services concurrently advantageously allows hardware resources to be allocated to services with a high degree of granularity.

FIG. **9** illustrates a preferred architecture for the servers **120** of the on-line services network **100**. All service DLLs **902** (two shown for the server **120** in FIG. **9**) run as part of a single, multi-threaded NT process **900**. The two service DLLs **902** shown in FIG. **9** could be, for example, a CHAT.DLL and a BBS.DLL, which implement the server portions of the CHAT and BBS service applications respectively. One or more threads **904** are process-wide threads which perform functions that are not specific to any particular service, while other threads **906**, **908** are dedicated to the particular services.

As is well known in the art, a thread is a schedulable entity of a process, and represents a sub-task of the process. During execution of a thread, the thread resides within the process's virtual address space. Advantageously, all of the threads of an NT process share the address space and the process's resources, including the process's access token, base priority, and object handles. A description of processes and threads under the Microsoft Windows NT operating system can be found in Helen Custer, *Inside Windows NT,* Microsoft Press, 1993.

In the preferred embodiment, a pool of threads is maintained for each server application running under the process **900**, and threads are dispatched from the pool on a per-service-request basis. In the example shown in FIG. **9**, a first pool **910** contains the threads **906** that are dedicated to SERVICE1.DLL, and a second pool **912** contains the threads that are dedicated to SERVICE2.DLL. Service requests (typically in the form of messages from client microcomputers **102**) that are routed to the server **120** are placed in a service request queue **914**, and are extracted from the queue **914** by the first available thread within the corresponding pool **910**, **912**. Although a single service request queue **914** is shown in FIG. **9**, a separate queue may be allocated to each service DLL.

The use of a single multi-threaded process (as opposed to multiple processes) to handle multiple sessions of the same service is particularly advantageous for services that permit the interaction or exchange of information between users. Examples of such services are CHAT and multi-user interactive games. With CHAT, for example, a message typed in by one user will be displayed on the monitors of one or more other users who are participating in the CHAT conference. Prior art designs, which use a separate process for each user, require interprocess communications (often between large numbers of processes) to accomplish this exchange of information between users, requiring a high degree of operating system overhead. Such overhead is significantly reduced in the present design, since no special data passing mechanisms are required for passing data between threads; one user-specific thread simply writes its data to the shared memory, and other user-specific threads then read the data as input.

A further advantage of using a single multi-threaded process to handle multiple service sessions is the ability to have individual threads that handle operations that are common to all users. For example, a single thread **904** of the process **900** can be used to monitor user inactivity for all users of the server **120**.

In accordance with another feature of the invention, particular services can be automatically activated or deactivated on the server **120** (or any server of the system) by loading and unloading service DLLs dynamically (i.e., without stopping the process **900**). In the preferred embodiment, the process **900** dynamically loads and unloads DLLs (in response to commands issued from a system console microcomputer over the LAN **122**) using the LoadLibrary(), GetProcAddress(), and FreeLibrary() functions of Microsoft's Win**32** API. The LoadLibrary() function is described at pages **86** and **87** of *Microsoft Win*32 *Programmer's Reference Manual, Volume* 4, (Microsoft Press, 1993,, ISBN 1-55615-518-2). The GetProcAddress() and FreeLibrary() functions are described at page 653, and at pages 484–485, respectively, of *Microsoft Win*32 *Programmer's Reference Manual, Volume* 3, (Microsoft Press, 1993, ISBN 1-55615-517-4). Briefly, these functions do the following. The LoadLibrary() function maps a specified DLL module (or other executable module) into the address space of the calling process **900**, and returns a handle to the module. The GetProcAddress() function uses the handle (returned by the LoadLibrary() function) and a DLL function name as input parameters, and returns the address of the specified DLL function. The FreeLibrary() unmaps a specified DLL module from the address space of the calling process **900**.

By way of example, to add the CHAT service to the server **120** shown in FIG. **9**, the DLL for the CHAT service (CHAT.DLL) would be loaded into RAM from the server's hard disk **918**. The process **900** would then call the LoadLibrary() and GetProcAddress() functions to link to CHAT.DLL and implement the CHAT service. The process **900** may subsequently use the FreeLibrary() function to dynamically unload the DLL. In the preferred embodiment, all service DLLs for all on-line services are stored on the hard disks **918** of all servers **120** of the system, so that any service can be automatically activated on any server **120** at any time.

Prior to dynamically unloading a service DLL from a server **120**, all sessions of the service (if any) must be moved to another server that is handling the same service. In the preferred embodiment, a "hot redirection" technique is used to transfer service sessions. The hot redirection technique works as follows. Initially, the Gateway **126** that is handling the logon session submits a serialization request to the server **120** that is handling the service session. The server **120** responds to the serialization request by returning an object that describes the internal state of the service session. This object is in the form of a byte stream which is normally meaningful only to a specific service. By way of example, the object for the hot redirection of a CHAT service session would normally indicate the conference name, the user name, the status of the user (e.g., a spectator, moderator, etc.), and the message number of the last CHAT message sent to the user in the CHAT conference. While the server is responding to the serialization request, the Gateway **126** buffers any service requests that are received from the client microcomputer **102**. The Gateway **126** then forwards the internal state information and the buffered service requests (if any) to a new server **120** (using the load balancing procedure of FIG. **6** to select a new server), and the new server resumes processing of service requests where the former server left off.

Once all service sessions have been transferred in this manner, the service DLL can be dynamically unloaded. A new service DLL can then be dynamically loaded to activate a different service, or to activate an updated version of the removed service.

The above-described hot redirection technique improves service reliability from the perspective of the end user by allowing service sessions to continue when corresponding servers **120** are taken down for maintenance. It is contemplated that this technique will be combined with a fault detection mechanism to automatically transfer service sessions to different servers when signs of abnormal service behavior are present.

7. Client-Gateway Communications

Communications between client microcomputers **102** and Gateways **126** are handled primarily by the Microsoft Connection Protocol (MCP), which is embodied within the MCP layers **208** (FIGS. **2**, **5**A and **5**B) in every client microcomputer **102** and every Gateway **126**. With reference to FIG. **10**, the MCP layer **208** comprises two protocol layers, a message layer **1000** and a packet layer **1010**. The message layer **1000** tags each outgoing message with a message length, a message type, and a service session identifier. In accordance with one feature of the invention, the message layer **1000** additionally multiplexes message streams, each message stream associated with a different service session. In accordance with another feature of the invention, when two or more message streams are multiplexed, the message layer **1000** allocates WAN bandwidth to the message streams based on predetermined (or user-specified) service priority levels.

The message layer **1000** uses the packet layer **1010** to provide a reliable byte stream between the client microcomputer **102** and the Gateway **126**. The packet layer **1010** handles error correction using CCITT CRC-**32**, and uses a sliding window protocol to handle flow control. To implement the sliding window protocol, each outbound packet contains an 8-bit packet sequence number which is incremented by the sender with successive packet transmissions. Each packet also contains the 8-bit packet sequence number of the first non-received packet (with erroneous packets treated as non-received packets), so that packets sent in one direction serve as acknowledgements for packets sent in the opposite direction. Special acknowledgement packets (i.e., packets that do not contain client-server message data) are transmitted after a timeout period if the receiver has no data to send. Sliding window protocols are well known in the art, and are described in Tannenbaum, *Computer Networks,* 2nd Ed., Prentice-Hall, Inc., 1989, pp. 223–239.

FIGS. **11**A and **11**B illustrate the preferred packet format used by the Microsoft Connection Protocol. Each packet **1100** has a length of either 256 bytes or 512 bytes, depending upon the identity of the telecommunications provider that provides the X.25 connection. Each packet **1100** includes a one-byte packet sequence number field **1102** followed by a one-byte acknowledgement number field **1104** (which indicates the sequence number of the first non-received packet). As indicated above, the packet sequence number and acknowledgement number are used to implement the sliding window protocol.

With the exception of special acknowledgement packets, each packet **1100** additionally includes one or more variable-length message segments **1106** (hereinafter "segments"). Segments are sub-units of messages, and are used by the MCP layer to multiplex message streams of different service sessions of a common end user. Messages are segmented on the originating side and reassembled at the receiving side. The first segment of a message specifies the length of the message. As further described below, the MCP layer allocates segments to service message streams in a round robin fashion, while varying the lengths of different segments in order to allocate different amounts of bandwidth to the different services.

With reference to FIG. **11**B, which illustrates the format of a segment **1110** (many of which may be contained in the segment field **1106** of FIG. **11**A), each segment **1110** includes a segment header **1112**, an optional one-byte segment length field **1114**, and a segment body **1116**. The segment header **1112** includes a multiplexing channel number (not shown). Multiplexing channels are allocated by the MCP layer to different service sessions (on a per-message-transmission basis) to allow the concurrent transmission of messages associated with different service sessions. In the preferred embodiment, a maximum of 16 simultaneous multiplexing channels can exist per user connection. When this multiplexing limit is reached, new message transmission requests are postponed until other messages are finished being transmitted. The limit of 16 multiplexing channels thus limits only the number of concurrent message transmissions, and not the number of service sessions.

Multiplexing channels may vary in bandwidth from one another. The MCP layer varies channel bandwidth by assigning different segment lengths to different service sessions. Segment lengths are preferably assigned on the Gateway side. The segment length assigned to a given service session can be modified (under the control of the service applications) during the lifetime of the service session. Segment lengths are preferably assigned when a service session is first opened (i.e., when the first segment of a new message stream is transmitted).

With further reference to FIG. **11**B, the segment header **1112** also includes three segment control bits (not shown). The first segment control bit is used to indicate whether or not the segment is the last segment of a message. The second segment control bit is used to indicate whether or not the segment extends to the end of the packet **1100**. When the segment does not extend to the end of the packet **1100**, the byte **1114** that follows the segment header **1112** will specify the number of bytes of message data contained within the segment body **1116**. The third segment control bit is used to indicate whether or not the message is a control message (such as a "close" request). When this third control bit indicates a control message, the first byte of the message indicates the control type.

Referring again to FIG. **11**A, each packet **1100** further contains a four-byte cyclical redundancy check (CRC) code **1120** (in accordance with CRC-32) to permit detection and correction of errors. Each packet **1100** ends with a packet delimiter **1122**, such as an ASCII carriage return code.

FIG. **12** illustrates, in example format, the preferred multiplexing and prioritization method of the present invention. A client microcomputer **102** is shown concurrently communicating with three different servers, a MAIL server, a CHAT server, and a VIDEO GAMES server (servers not shown). Within the client microcomputer **102**, the MCP layer has allocated portions of RAM to the respective services to form a MAIL message buffer **1202***a*, a CHAT message buffer **1204***a*, and a VIDEO GAMES message buffer **1206***a*. Likewise, within the Gateway **126** to which the client microcomputer **102** is connected, the MCP layer has allocated portions of RAM to form corresponding MAIL, CHAT and VIDEO GAMES message buffers **1202***b*, **1204***b* and **1206***b*. Each buffer in FIG. **12** is associated with a particular service session, and can be thought of as including two separate buffers, a first buffer for holding outgoing messages and a second buffer for receiving incoming messages.

A packet **1210** being sent from the Gateway **126** to the client microcomputer **102** contains segments **1212** corresponding to the MAIL, CHAT and VIDEO GAMES services. Each segment **1212** contains a portion of an outgoing message stored in one of the buffers **1202***b*, **1204***b*, **1206***b*. The packet **1210** is generated by the MCP layer by extracting and multiplexing message data from the buffers **1202***b*, **1204***b*, **1206***b* in a round robin fashion. When the packet **1210** arrives at the client microcomputer **102**, the MCP layer extracts and demultiplexes the segments, and stores the message data in the corresponding MAIL, CHAT and VIDEO GAMES buffers **1202***a*, **1204***a*, and **1206***a*. Messages sent from the client microcomputer **102** to the Gateway **126** are multiplexed, packetized, and demultiplexed in the same manner.

As recognized by those skilled in the art, different types of on-line services tend to have different throughput and/or latency demands. For example, an on-line interactive video games service will likely require a greater throughput for satisfactory operation than a mail service. Thus, with the multiplexing scheme of the present invention, it is desirable to implement a corresponding service priority scheme, whereby different services are allocated different amounts of the available WAN bandwidth. This is preferably accomplished by assigning segment lengths to different on-line services based on predetermined entries stored within a service priority table **1220**. The service priority table **1220** contains priority levels (in the form of maximum segment lengths) for each on-line service (or class of on-line service), and is accessed by the MCP layer whenever a new service is opened by the user. The service priority table may additionally include minimum throughput requirements for certain services. For example, for an embodiment of the CHAT service that permits voice communications, the table **1220** could indicate a minimum throughput requirement of 7 kbps (to ensure voice reproduction of a predetermined quality), and request to open the service could be failed when this amount of bandwidth is not available.

In the example shown in FIG. **12**, the MCP layer has allocated segment lengths of 40 bytes, 60 bytes and 80 bytes, respectively, to the MAIL, CHAT and VIDEO GAMES services. Thus, the MAIL service has been allocated half the bandwidth of the VIDEO GAMES service, and two-thirds the bandwidth of the CHAT service.

The service priority table **1220** may reside within either the client microcomputer **102**, the Gateway **126**, or both. Alternatively, the throughput priorities could be provided by the service applications, eliminating the need for the table **1220**. In the preferred embodiment, the throughput priorities indicated by the table **1220** cannot be overridden by end users. In other embodiments, client applications may be written to allow end users to override the predetermined throughput priorities.

Because throughput demands for client-to-server data transfers of a service may differ significantly from the throughput demands for server-to-client data transfers (for the same service), it may be desirable to use a first set of bandwidth allocations for client-to-Gateway message traffic, and second set of bandwidth allocations for Gateway-to-client message traffic. This may be accomplished by maintaining two priority levels for each service, one for client-to-server messages, and another for server-to-client messages.

Although the above-described multiplexing and bandwidth allocation schemes are described in the context of packets, it will be recognized that the packetization of message segments is not essential. For example, in embodiments which use TCP/IP for client-Gateway communications, the variable-length segments are simply multiplexed within the outgoing byte stream, and no pack-

etization by the client microcomputer **102** or the Gateway **126** is necessary.

8. Arbiter

As indicated above, certain types of services make use of the Arbiter service to replicate update transactions. In the preferred embodiment, the Arbiter is implemented by dedicated Arbiter microcomputers **128** (shown in FIGS. **1** and **3**) that are assigned to particular service groups. Advantageously, the Arbiter software is service-application-dependent, and thus does not have to be customized for each service. Service applications that require the Arbiter are simply written to make use of the services provided by the Arbiter, as further described below.

The Arbiter performs data set replication as follows. Whenever a server **120** of a service group receives a request (typically from a client microcomputer **102**) that requires an update to the service's content data, the server generates and sends an update transaction to the service group's Arbiter microcomputer **128** (assuming the service group uses the Arbiter service) without processing the transaction. This transaction specifies the data entity (such as a BBS file) to be updated, and the specific update to be made to the data entity. The Arbiter microcomputer **128** records each update transaction it receives in its transaction log, and dispatches each update transaction to every server **120** in the service group (by issuing remote procedure calls to the servers **120**), including the server **120** that generated the transaction. (Each Arbiter microcomputer **128** keeps track of the servers **120** within its respective service group by monitoring the global registry **145**, which includes a mapping of servers to service groups).

The servers **120** that receive the update transaction from the Arbiter respond by processing the update transaction, and by returning a status code. The update transactions are dispatched to the servers **120** in a serialized order (preferably on a first-submitted-first-dispatched basis), so that all servers of the service group process the update transactions in the same order.

Each time an update transaction is dispatched by the Arbiter, the Arbiter monitors the outcome (typically success or failure) of the transaction on each server **120** by checking the status codes returned by the servers **120**. When one server **120** of the service group processes the dispatched transaction differently than the other servers, the Arbiter takes that server off-line. More generally, if a minority number of servers **120** of the service group report a different outcome than the other servers, the minority servers are treated as being "wrong" and are taken off-line. The Arbiter service thereby ensures that all operational servers **120** within the service group contain consistent service content data.

By way of example, suppose that the BBS server **120***f* (FIG. **3**) receives a user service request that will cause a BBS message to be added to a specified folder, and that the BBS server **120***g* receives a service request that will cause another BBS folder to be deleted. Since both of these requests require updates to BBS content data, the servers **120***f* and **120***g* send update transactions to the BBS Arbiter microcomputer **128** (without updating their own copies of the BBS content data). (Note that it is the responsibility of the service applications to detect and forward update transactions.) The BBS Arbiter microcomputer **128** (hereinafter "BBS Arbiter **128**") records these transactions in its transaction log **306**. One or more dispatcher threads (not shown) running on the BBS Arbiter **128** read the update transactions from the transaction log, and dispatch the update transactions (in the order received) to each of the BBS servers **120***e–g*.

Upon processing each update transaction, the servers **120***e*–**102***g* report back to the BBS Arbiter **128** on the success or failure of the transaction by returning a status code. To provide a specific example, if the servers **120***e* and **120***f* report successful completion of the update transaction and the server **120***g* reports a failure, the "odd-ball" server **120***g* is taken off-line by the BBS Arbiter **128**. If, on the other hand, the servers **120***e* and **120***f* report a failure of the transaction and the server **120***g* reports a success, the server **120***g* is again taken off-line (since it represents the minority outcome), even though the server **120***g* reported a successful completion of the transaction. Whenever a server **120** is taken off-line, a message is sent to a system console (not shown) to inform a human operator of the problem.

For each update transaction, the Arbiter microcomputer **128** records in the transaction log the following information: the time of the transaction, a transaction ID (TRID), the originator of the transaction, the data set (or data sets) affected by the transaction, the contents of the transaction (contents dependent upon the particular service), the transaction results received from the servers **120**, and the status of the transaction (pending, success, failure, etc.). The Arbiter transaction log is stored in non-volatile memory, and continues to grow in size until trimmed. Transaction logs are trimmed according to trimming policies which are specified in the global registry **145**.

Advantageously, the transaction log of an Arbiter microcomputer **128** typically contains all of the information needed to automatically bring up a new server within the corresponding service group. The new server **120** being brought up may, for example, be a server that was taken off-line (with respect to the service group) for maintenance, or a server that is being reallocated to the service group from a lightly loaded service group, or a server that is being added to the host data center **104**.

When a new server is added to a service group (i.e., when the appropriate service DLL or DLLs are initially loaded on the new server), the Arbiter issues the update transactions stored in the corresponding transaction log to the new server **120** while monitoring the outcome of each transaction. The transactions are submitted to the new server **120** in the same order as previously dispatched to the other servers **120** in the service group, so that the new server is brought to the same state as the existing servers. If the transaction log does not go back far enough to bring up the new server **120**, a message is sent to the system console, and a backup of the service group's transaction log is used to supplement the current transaction log.

During this process of synchronizing or "rolling forward" the new server **120**, the service application running on the server remains in an "invalid" state, and no service requests from client applications are taken. Once all of the transactions recorded within the transaction log have been processed by the new server **120**, the state of the new server is set by the Arbiter to "active," and the new server begins receiving service requests.

Although the Arbiter service is preferably implemented on service-specific Arbiter microcomputers which run the Arbiter software, it will be recognized that the functionality of the Arbiters could be implemented elsewhere within in the host data center **104**. For example, Arbiter functionality could be implemented by a server **120** of a service group. It will also be recognized that two or more service groups could share the same Arbiter microcomputers **128**, and that multiple Arbiter microcomputers **128** could be assigned to the same service group to handle different types of transactions.

A preferred implementation of the Arbiter service is described in further detail in a concurrently filed patent application having the title TRANSACTION REPLICATION SYSTEM AND METHOD FOR SUPPORTING REPLICATED TRANSACTION-BASED SERVICES, which is incorporated herein by reference.

9. Dynamic Allocation of Servers to Service Groups

Loads placed on particular services (or equivalently, particular service groups) may fluctuate relative to one another on a daily basis due to fluctuations in usage of different services. Usage of a STOCK QUOTE service, for example, may peek during regular business hours, while usage of a CHAT service may peak at night or on weekends. To accommodate for such fluctuations in service usage levels, the on-line services network **100** allocates servers **120** to service groups based on service loads.

In a preferred embodiment, the on-line services network **100** maintains a pool of unused servers **120**. Servers **120** are taken from the pool and allocated to service groups when high service group loads (i.e., high service loads) are detected, and are added to the pool when low service group loads are detected. Servers may additionally or alternatively be reallocated from service group to service group without being added to the pool. Because overall usage levels of on-line services networks typically fluctuate with time (with, for example, peak usage occurring during daytime and early evening hours, and lighter usage levels occurring during early morning hours), the size of the pool will typically fluctuate significantly on a daily basis.

The framework for being able to dynamically allocate servers **120** to service groups based on service group loads is set by the service map, dynamic DLL loading/unloading, hot redirection, and Arbiter features discussed above. Specifically, the service map **136** indicates the loads of all operational servers **120**, and thus enables the loads of the various service groups to be determined; dynamic loading and unloading of service DLLs enables different service applications to be activated and deactivated on different servers, and thus enables servers **120** to be automatically transferred from service group to service group (or between a pool of unused servers and a service group); hot redirection enables a service session to be transferred from one server to another, so that a server can be removed from a service group without prematurely terminating a service session; and the Arbiter **128** enables new servers **120** to be "rolled forward" to the same state as the other servers of a service group (for services that use dynamic, replicated data sets). Thus, with the addition of a mechanism for monitoring service group loads, servers **120** can be dynamically allocated to service groups to achieve a balanced allocation of processing resources to services.

In one embodiment of the on-line services network **100**, service group loads are monitored by a system operator who decides when and how to allocate servers **120** to service groups. In another embodiment, the process of allocating servers **120** to service groups is completely automated, with a load monitoring program (which runs, for example, on one of the administrative servers **134**) deciding when and how to reallocate servers. A load monitoring program of this type is illustrated in FIG. **13**.

FIG. **13** is a high level flow chart of a preferred routine for monitoring the load on a particular service, service i, and for determining whether a server **120** should be added to or extracted from the corresponding service group, service group i. The routine is preferably run continuously (on a service-by-service basis) on a dedicated administrative server **134** that receives the periodically broadcasted service map **136**. With reference to blocks **1300** and **1302**, the service map **136** is initially accessed to identify the servers **120** currently allocated to the service group, and to read the CPU LOAD values of each identified server. These CPU LOAD values are then averaged to obtain the load of the service group, SRVGRP_LOAD. With reference to block **1304**, the SRVGRP_LOAD values calculated over the last N service maps are then averaged to obtain the average load of the service group, AVG_SRVGRP_LOAD, over a predetermined period of time (defined by N). The effect of this step is to filter out short-term fluctuations in the load of the service group, so that servers **120** will not unnecessarily be taken from or added to the service group. The variable N can be set to different values for different services.

With reference to blocks **1306** and **1308**, the average load AVG_SRVGRP_LOAD is then compared to a predetermined minimum, MIN_LOAD, to determine whether a server **120** should be removed from the service group. If the average load is less than the predetermined minimum, a server **120** will be extracted from the service group (assuming there are currently at least two servers in the service group) and added to the pool of unused servers. With reference to blocks **1312** and **1314**, the average load is also compared to a predetermined maximum, MAX_LOAD, to determine whether or not to add a server **120** to the service group. If the average load is greater than the predetermined maximum, a server **120** will be added to the service group from the pool (assuming there is currently at least one server in the pool). The predetermined MIN_LOAD and MAX_LOAD values can be set by a system operator, and can be set to different values for different services. An appropriate algorithm could be used to adjust the MIN_LOAD and MAX_LOAD values for a service based on the number of servers **120** currently allocated to the service group.

As indicated above, whenever a server **120** is transferred to or from a service group, the change is reflected in the service map **136**, causing the Gateways **126** to route service requests in accordance with the change. For example, when a server **120** is added to a MAIL service group, the server will inform the service map dispatcher **144** of its change in status (by transmitting its local map **140** to the service map dispatcher **144**), and the service map dispatcher **144** will include the status change (along with the CPU LOAD and CPU INDEX values for the server) in the next service map **136**. Once the updated service map **136** has been broadcasted to the Gateways **126**, the Gateways will route "open" requests to the newly added server in accordance with an appropriate load balancing method (as described above) . Likewise, when a server **120** is extracted from a service group and added to the pool of unused servers, the server's local map **140** will not appear in the next service map **136**, and the Gateways **126** will stop routing service requests to the removed server.

10. Scalability

An important aspect of the on-line services network **100** described above is its ability to be scaled in capacity without having to change the existing system and application software, and without having to reconfigure existing network components. In accordance with the architecture of the system, new service applications can (and normally will) be written to interact with existing Gateway **126**, Arbiter **128** and service map dispatcher **144** software components so that new services can be added without changing these components. By way of example, because the Arbiter **128** is implemented as a generic (i.e., service-independent) transaction replication service, new service applications can be written to make use of the services provided by the Arbiter

128, and no changes to the existing Arbiter software will be necessitated by the addition of such service applications.

Scaling of system capacity is also facilitated by the service map feature of the architecture. For example, new servers **120** and/or new service applications can be added to the system without having to manually update service request routing tables or Arbiter tables.

Scalability is also provided by the ability to allocate a variable number of replicated servers to a given service; increased demand for a service can be accommodated for by simply allocating additional servers to the popular service. Because the replicated servers of a service group handle service requests independently from one another, no performance bottlenecks exist; thus, increases in service loads can be accommodated for (by the addition of servers) without any significant degradation in performance.

Although this invention has been described in terms of certain preferred embodiments, other embodiments that are apparent to those or ordinary skill in the art are also within the scope of this invention. Further, although the invention has been described in the context of on-line services networks, it will be recognized that certain aspects of the invention are applicable to other types of computer networks. By way of specific example, the dynamic load balancing methods described herein could be used to select between servers of a distributed data base system. Accordingly, the scope of the present invention is intended to be defined only by reference to the appended claims.

What is claimed is:

1. A data center for providing on-line service to end users who connect to the data center from client computers that run client applications, comprising:

a plurality of application servers interconnected by a local area network, each application server running at least one service application that implements an on-line service, different groups of said application servers running different service applications to provide different on-line services, application servers which run like service applications forming service groups; and

a plurality of gateway computers connected to said application servers by said local area network and connected to the client computers by a wide area network, each gateway computer locally storing a respective copy of a service map that includes a mapping of application servers to service groups, said service map automatically updated by said plurality of application servers, each gateway computer programmed to receive service requests from client computers and to route said service requests to application servers running corresponding service applications, the gateway computers further programmed to establish and maintain bi-directional data connections between the client computers and said application servers over the local area network based on entries within the service map, the data connections corresponding to user service sessions.

2. The data center according to claim **1**, each gateway computer further programmed to, upon receiving a service request from a client computer to open a user-specified service, doing each of:

(a) identifying all application servers that fall within a service group which corresponds to said user-specified service,

(b) determining respective processing loads for each application server identified in step (a), and

(c) based on processing loads determined in step (b), selecting an application server from said service group to process said request to open said user-specified service.

3. The data center according to claim **2**, wherein step (b) comprises accessing said service map, said service map indicating processing loads for substantially all of said application servers, including application servers of multiple different service groups, said service map automatically updated by said application servers to reflect changes in processing loads.

4. The data center according to claim **2**, wherein step (c) comprises selecting an application server with a greatest available CPU processing power.

5. The data center according to claim **2**, wherein step (c) comprises selecting an application server such that a probability of selection is proportional to an available CPU processing power.

6. The data center according to claim **1**, each gateway computer further programmed to, upon receiving a service request from a client computer to open a user-specified service, doing each of:

(a) calling a service-specific locator to identify a unique one of said application servers that corresponds to said service request, and

(b) routing said service request to the application server identified in step (a).

7. The data center according to claim **6**, wherein step (a) comprises using a user identifier to identify one of a plurality of MAIL servers.

8. The data center according to claim **6**, wherein step (a) comprises using a user identifier and a service identifier to identify said unique application server.

9. The data center according to claim **6**, wherein step (a) comprises using a conference locator to identify one of a plurality of CHAT servers.

10. The data center according to claim **9**, wherein step (a) comprises identifying a CHAT server which is assigned to a user-specified CHAT conference.

11. The data center according to claim **1**, further comprising a service map dispatcher connected to said local area network, said service map dispatcher programmed to receive information from said application servers and to generate said service map therefrom, said service map dispatcher further programmed to periodically broadcast an updated version of said service map to said gateway computers.

12. The data center according to claim **11**, wherein said service map contains a mapping of application servers to service applications, said service map thereby enabling said gateway computers to identify application servers to handle specific service requests.

13. The data center according to claim **11**, wherein said service map contains, for each of said application servers, a client count value that indicates a current number of service sessions.

14. The data center according to claim **11**, wherein said service map contains, for each of said application servers, a processing load value which indicates a current application server processing load.

15. The data center according to claim **14**, wherein said service map additionally contains, for each of said application servers, a CPU index value which indicates an overall application server processing power.

16. The data center according to claim **1**, each gateway computer further programmed to receive a multiplexed data stream from a client computer, said multiplexed data stream containing alternating portions of different messages, at least two of said different messages addressed to different services, each gateway computer further programmed to demultiplex said multiplexed data stream to reconstruct said messages, and to route reconstructed messages to corre-

sponding application servers, said gateway computers thereby permitting a client computer to concurrently communicate with multiple application servers so that an end user can simultaneously access multiple on-line services via a single gateway computer.

17. The data center according to claim **16**, wherein said portions of different messages are varied in size so as to allocate different amounts of wide area network bandwidth to different services.

18. The data center according to claim **1**, each gateway computer further programmed to receive multiple data streams from said application servers and to multiplex said data streams for transmission to a client computer, each of said data streams generated by a different application server and corresponding to a different on-line service, said gateways thereby allowing an end user to concurrently receive data from multiple on-line services.

19. The data center according to claim **18**, wherein each gateway multiplexes said data streams by combining variable-length portions of said data streams, lengths of said variable-length portions being varied from on-line service to on-line service to allocate different amounts of bandwidth to different services.

20. The data center according to claim **1**, further comprising a load monitoring computer programmed to monitor loads of said service groups, and to add an application server to a service group when a load on said service group exceeds a predetermined threshold.

21. The data center according to claim **1**, wherein at least one of said application servers comprises a hard drive having stored thereon a plurality of dynamically loadable service modules, each dynamically loadable service module comprising executable code of a respective service application, said dynamically loadable service modules enabling an application server to be automatically allocated from one service to another, the at least one application server further including code for transferring a user service session to another application server when the application server is reallocated.

22. The data center according to claim **21**, wherein the at least one application server runs a process that dynamically loads and unloads different executable service modules to activate and deactivate different service applications.

23. The data center according to claim **22**, wherein said process is a multi-threaded process.

24. The data center according to claim **1**, wherein each application server of at least one service group locally stores a replicated copy of on-line service data, said on-line service data provided to end users of an on-line service that corresponds to said service group.

25. The data center according to claim **1**, further comprising a client microcomputer configured to access said on-line services, the client microcomputer comprising:

a plurality of client applications, each client application configured to communicate with a corresponding server application of the data center to thereby implement an on-line service, each client application generating an outgoing data stream to a corresponding server application and receiving an incoming data stream from said corresponding server applications; and

a communications program configured to multiplex multiple of said outgoing data streams for transmission to the data center, and configured to separate multiplexed incoming data streams and route said incoming data streams to corresponding of said client applications, said communications program thereby enabling a user of the data center to concurrently access multiple on-line services.

26. The client microcomputer according to claim **25**, said communications program further configured to multiplex said outgoing data streams by combining variable-length data segments from alternating of said outgoing data streams, lengths of said variable-length segments assigned by said communications program according to throughput priorities of corresponding on-line services.

27. The client microcomputer according to claim **25**, said communications program further configured to combine said outgoing data streams within packets, each packet containing segments from alternating of said outgoing data streams.

28. The client microcomputer according to claim **27**, wherein said segments are variable-length segments, and wherein lengths of said segments are assigned on a per-client application basis.

29. The client microcomputer according to claim **27**, said communications program further configured to implement a sliding window flow control protocol.

30. The client microcomputer according to claim **27**, said communications program further configured to implement CCITT CRC-**32** error correction.

31. In a computer network which comprises a gateway computer that connects a wide area network to a local area network, a method of processing service requests from users so as to balance a processing load among application servers of the network, the application servers interconnected by the local area network, the method comprising the steps of:

receiving a service request at a host data center, said service request transmitted from client computer to said host data center over said wide area network, said service request corresponding to a service;

identifying a group of application servers that are assigned to said service, said step of identifying comprising accessing a service map stored by said gateway computer, said service map automatically updated by said application servers;

determining processing loads of each application server in said group;

based on said processing loads, selecting an application server from said group to process said service request; and

routing said service request to a selected application server.

32. The method according to claim **31**, wherein said step of selecting comprises calculating an available CPU processing power for each application server in said group.

33. The method according to claim **32**, wherein said step of selecting further comprises selecting an application server having the greatest available CPU processing power.

34. The method according to claim **32**, wherein a probability of selection of an application server is proportional to the available CPU processing power for said application server.

35. The method according to claim **32**, wherein said available CPU processing power is calculated as a function of a CPU load value and a CPU index value for each respective application server of said group, said CPU load values indicating loads on respective application servers caused by service requests, said CPU index values indicating overall processing powers of respective application servers.

36. The method according to claim **31**, wherein said step of determining comprises accessing said service map, said service map indicating processing loads for said application servers.

37. The method according to claim 36, wherein said service map is automatically updated on a substantially periodic basis to reflect changes in application server loads.

38. The method according to claim 36, wherein said service map is automatically updated on an average of at least once per minute to reflect changes in application server loads.

39. The method according to claim 37, wherein said service map is broadcasted to a plurality of service request routers of said host data center on a substantially periodic basis.

40. The method according to claim 31, wherein said service map includes a mapping of application servers to services for substantially every application server of said host data center.

41. The method according to claim 31, wherein said step of receiving, identifying, determining and selecting are performed by said gateway computer.

42. The method according to claim 31, wherein said service request comprises a request to open service.

43. The method of claim 31, further comprising the steps of:

generating a plurality of data streams between said client computer and said gateway computer, each data stream corresponding to a respective service session, each service session handled by a different application server; and

multiplexing said data stream for concurrent transmission over said wide area network;

the method thereby enabling concurrent use of multiple services by a single user.

44. The method according to claim 43, further comprising, at said gateway computer, doing each of:

demultiplexing multiplexed data streams generated by said client computer, and

forwarding demultiplexed data streams to corresponding application servers.

45. The method according to claim 43, wherein said step of multiplexing comprises allocating different quantities of wide area network bandwidth to different data streams of said plurality of data streams to thereby accommodate for differing throughput priorities of different services.

46. The method according to claim 43, wherein said step of multiplexing comprises generating a packet for transmission over said wide area network, said packet containing segments of each data stream of said plurality of data streams.

47. The method according to claim 46, wherein said segments are variable length segments, and wherein lengths of said segments are varied from data stream to data stream in order to allocate differing quantities of available wide area network bandwidth to different services.

48. The method according to claim 47, wherein said lengths are varied in proportion to predetermined throughput priorities of corresponding services.

49. The method according to claim 47, wherein said lengths are varied in proportion to user-specified throughput priorities of corresponding services.

50. A gateway computer, comprising:

a first network interface for communicating with client computers;

second network interface for communicating with a plurality of servers;

a memory, said memory containing load data for said plurality of servers, said load data updated periodically such that said load data indicates current loads for said plurality of servers, said load data indicating, for each server of said plurality of servers, an available processing power for processing requests from client computers; and

a CPU, said CPU programmed to each of:

(a) receive a request from a client computer,

(b) in response to said request, access said load data to identify a server of said plurality of servers that is lightly load, and

(c) route said request to said server identified in step (b).

51. The gateway computer according to claim 50, wherein said first network interface is configured for communications over a wide area network, and wherein said second network interface is configured for communications over a local area network.

52. The gateway computer according to claim 50, wherein said memory further comprises application mapping data, said application mapping data identifying server applications running on different servers of said plurality of servers.

53. The gateway computer according to claim 52, said CPU further programmed to, upon receiving a request which identifies an application, accessing said application mapping data to identify a group of servers of said plurality of servers that is running said application.

54. The gateway computer according to claim 50, said CPU further programmed to:

(d) receive respective data streams from at least two servers of said plurality of servers, and

(e) multiplex said data streams for transmission to said client computer.

55. A method of disseminating load data of a plurality of servers to a plurality of routing computers that route client requests to the servers based on the load data received from a dispatcher entity, the servers, the routing computers and the dispatcher entity interconnected by a local area network, the method comprising the steps of:

(a) at each server of the plurality of servers, automatically determining a current load on the respective server, and automatically transmitting load data that reflects the current load to the dispatcher entity of the local area network; and

(b) at the dispatcher entity, automatically compiling the load data received from each of the plurality of servers into a map, and broadcasting the map on the local area network to allow the routing computers to obtain the load data of the plurality of servers.

56. The method of claim 55, wherein step (a) comprises determining an available processing power of each server for handling client requests.

57. The method of claim 55, wherein step (a) comprises, at each server of the plurality of servers, sending service allocation data to the dispatcher entity, the service allocation data indicating at least one service to which the respective server is currently allocated, and wherein step (b) comprises broadcasting the allocation data of the plurality of servers from the dispatcher entity on the local area network.

58. A data center for providing on-line services to end users who connect to the data center from client computers that run client applications, comprising:

a plurality of application servers interconnected by a local area network, each application server running at least one service application that implements an on-line service, different groups of said application servers running different service applications to provide different on-line services, application servers which run like service applications forming service groups; and

a plurality of gateway computers connected to said application servers by said local area network and connected to the client computers by a wide area network, each gateway computer programmed to receive service requests from client computers and to route said service requests to application servers running corresponding service applications, the gateway computers further programmed to establish and maintain bi-directional data connections between the client computers and said application servers over the local area network based on entries within the service map, the data connections corresponding to user service sessions, each gateway computer further programmed to, upon a service request from a client computer to open a user-specified service, doing each of:

(a) identifying all application servers that fall within a service group which correspond to said user-specified service,

(b) determining respective processing loads for each application server identified in step (a), the step of determining comprising accessing a service map, said service map indicating processing loads for substantially all of said application servers, including application servers of multiple different service groups, said service map automatically updated by said application servers to reflect changes in processing loads, and

(c) based on processing loads determined in step (b), selecting an application server from said service group to process said request to open said user-specified service.

59. The method of claim **55**, wherein steps (a) and (b) are performed periodically.

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UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : 5,774,668
DATED : June 30, 1998
INVENTOR(S) : Choquier et al.

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In Column 12, line 33: "Requests" should begin a new paragraph.

IN THE CLAIMS:

In Claim 50, Column 29, line 63: "second" should read --a second--.

Signed and Sealed this

Twenty-fourth Day of August, 1999

Attest:

Q. TODD DICKINSON

Attesting Officer

Acting Commissioner of Patents and Trademarks